

Petroleum Subsidy Reform According to HANK: Case of Indonesia

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An Advanced Research Essay submitted for the degree of

Master of Economics

The Australian National University

Research School of Economics



**Australian
National
University**

May 2026

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To my late father (Buya) and my late mother (Umi Yul), who devoted their lives to their children.

Disclaimer

I hereby declare that the work undertaken in this thesis has been conducted by me alone, except where indicated in the text. I conducted this work between July 2025 and May 2026, during which period I was enrolled in ECON8887 Advanced Research Essay as Master's student in Economics at the Research School of Economics, College of Business and Economics, the Australian National University. This thesis, in whole or any part of it, has not been submitted to this or any other university for a degree.

A handwritten signature in black ink, appearing to read 'Syahid Izzulhaq', with a long, sweeping horizontal stroke extending to the right.

Syahid Izzulhaq
May 27, 2026

Acknowledgment

I would like to thank my supervisor Associate Professor Chung Tran for his support, guidance, and useful feedback and Indonesian Endowment Fund for Education (LPDP) for providing a fully funded scholarship.

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Petroleum Subsidy Reform According to HANK: Case of Indonesia

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Abstract

This paper studies the economic implications of petroleum subsidies in Indonesia using a Heterogeneous Agent New Keynesian (HANK) model calibrated to Indonesian economy. The results suggest that petroleum subsidy in Indonesia generates suboptimal outcomes relative to alternative policy regimes. Specifically, I find that eliminating petroleum subsidies increases output, improves welfare, and promotes a more equal distribution of income and wealth in the long run. Transfer-based policies, especially targeted transfers, further enhance welfare and equality outcomes. Despite these long run gains, this paper shows that policy reforms are associated with short-run economic costs such as output contraction, inflationary pressure, and widening fiscal deficits. As stabilisation instrument, I find that petroleum subsidy is ineffective. Although it mitigates initial price surges, it leads to larger output losses, more persistent price increases, and greater fiscal deterioration in response to oil price shocks. The petroleum subsidy also performs poorly to protect the poor from adverse oil price shocks.

Keywords: Petroleum subsidies, Energy reform, HANK, DSGE, Indonesia.

JEL Classifications: D31, E32, E62, Q43, Q48.

Word Count: 8,618 Words.

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1 Introduction

Supply shocks, particularly energy price shocks, have been a defining feature of the global economic cycle over the past decade. Since monetary policy has limited capacity to respond to such shocks (Fornaro and Wolf, 2023), the attention has been directed toward the role of fiscal policy, such as energy subsidies, as a stabilisation instrument. However, studies on the role of fiscal policy as energy price shock absorber remain limited, especially for the case of developing countries. This paper studies economic implications of petroleum subsidy policy in Indonesia.

First introduced in the late 1970s, petroleum subsidy has remained a central fiscal instrument in Indonesia to primarily protect low income households from petroleum price volatility and to preserve macroeconomic stability. The subsidies enable all consumers to purchase certain petroleum products at subsidised price, which is below the market price, with the Government of Indonesia (GoI) covering the price differentials. In practice, however, the subsidised petroleum prices are relatively rigid and do not automatically adjust to global price fluctuations. This imposes a substantial fiscal burden. For instance, during the global energy shocks in 2022, petroleum subsidies in Indonesia accounted for approximately 14% of total government expenditure, 12 percentage points higher than in 2020.

Against this background, this paper evaluates the capacity of current petroleum subsidies in Indonesia (*ad hoc* price adjustment or baseline policy regime hereafter) by comparing it to alternative fiscal policy regimes including **no subsidies, automatic price adjustment, targeted transfers, and (universal) lump sum transfers**, along three dimensions. First, it evaluates the long run macroeconomic, fiscal, and equity-related implications. Second, this paper analyses the transitional consequences of subsidy reform. Lastly, this paper assesses which policy regimes can provide the best balance between macroeconomic stabilisation and fiscal sustainability.

To address these questions, I construct Heterogeneous Agent New Keynesian (HANK) model calibrated to the Indonesian economy. HANK has been a cornerstone of the modern economic modelling and offers numerous advantages in the study of fiscal policy as it captures households household heterogeneity. Unlike Representative Agent New Keynesian (RANK) models, which imply Ricardian Equivalence, HANK resembles Keynesian Cross, thereby providing more realistic effects of fiscal policy (Moll, 2024; Auclert et al., 2024).

The model developed in this paper is motivated by Pieroni (2023). The economy is

consisted of heterogeneous households, union (and recruiting firm), intermediate goods producers, final goods producers, government (fiscal policy), and central bank (monetary policy). The households are ex-ante identical but ex-post heterogeneous due to idiosyncratic productivity shocks and incomplete market. Both nominal wage and price level are sticky. Capital is abstracted from the model. Different from Pieroni (2023), I model the oil prices as exogenous. This essentially means that Indonesian economy is price taker in global oil markets in which aligns with the fact that Indonesia is oil-importing country. On the fiscal side, I treat consumption tax rate fixed, while labour income taxes adjust according to a fiscal rule to ensure long-term debt solvency. This assumption is broadly consistent with Indonesia’s historical tax policy where consumption tax is barely changing throughout decades, only adjusted once in 2023 since 1983.

The results suggest that baseline policy regime results in suboptimal outcomes compared to the alternatives. In the long run, removing all petroleum subsidies increases output, improves welfare, promotes a more equal distribution of income and wealth, and strengthens fiscal positions albeit at the expense of higher price level. Providing transfers can further improve welfare and equality in the long run, although they cause a slightly lower output compared to no subsidies. Despite these long run gains, this paper shows that reforms are challenging as they are associated with economic contraction and widening fiscal deficits during transitions. Such unpleasant short run consequences of a reform may rationalise why is government reluctant to implement reforms.

In regard to stabilisation, I find that baseline regime is counterproductive. Although it mitigates initial price surges, it also causes larger the output losses and more persistent price increases. Primary balance also strongly deteriorates under the baseline policy regime compared to other policy regimes. Furthermore, this paper also finds that the baseline regime also performs worse to protect the poor from the shocks, in particular because they experience a larger decline in consumption in response to oil price shocks than richer households.

Related Literature. This paper is closely related to the literature on the role of fiscal policy in presence of energy shocks. This strand of literature have been growing, especially after energy price shocks in 2022. However, these studies primarily focus on the case of developed countries. For instance, Pieroni (2023) and Glocker and Wegmüller (2024) calibrated their model using European countries while Auclert et al. (2023) uses the US economy and Erceg et al. (2024) for the both the US and Euro Area economies. In contrast, this paper is tailored to capture the characteristics of

developing economy, in particular Indonesia.

Several works have examined the role of energy subsidies in developing countries, although still limited. Ginn (2024) studies the petroleum subsidies in Malaysia and Plante (2014) in several developing countries. However, Ginn (2024) only focuses on two types of fiscal instruments which are universal and targeted subsidies while Plante (2014) only examines the long run macroeconomic impacts of fuel subsidies. On the contrary, this paper considers a broader set of policy regimes and focuses not only on long run implications but also on short run stabilisation dynamics and transitional effects of reform.

Furthermore, this paper is also closely related to Indonesian case studies. Clements et al. (2003) and Yusuf et al. (2017) investigate the real and distributional impact of petroleum price liberalisation in Indonesia using Computational General Equilibrium Model (CGE). Dartanto (2013) examines the implication of eliminating petroleum subsidies on fiscal balance. While these studies have a quiet similar objective, this paper extends the analysis by examining not only the long run outcomes but also the transitional effects of policy reforms and the short run stabilisation dynamics. This paper also considers a broader fiscal policy regimes. In addition, it is also the first to employ a HANK model to analyse petroleum subsidies in the Indonesian context.

This paper is organised as the following sections. Section 2 elaborates on institutional background. Section 3 discusses the economic model I use in this paper. Lastly, Section 4 and 5 are results and conclusion, respectively.

2 Institutional Background

Petroleum subsidies in Indonesia was first introduced in the late 1970s in response to global oil price escalation. During that period, the GoI set retail price for various petroleum products including aviation turbine fuel (avtur) and gasoline (avgas), gasoline (RON 88), kerosene, and diesel, and covered the price gaps. The policy was aimed to protect consumers from volatile global oil price while promoting industrialisation (Yusuf et al., 2017; Jazuli et al., 2021). Between 1980 to the late 1990s, the Government of Indonesia (GoI) adjusted retail price infrequently. Although these policies imposed substantial fiscal costs, the GoI can maintain sufficiently high oil revenue.¹

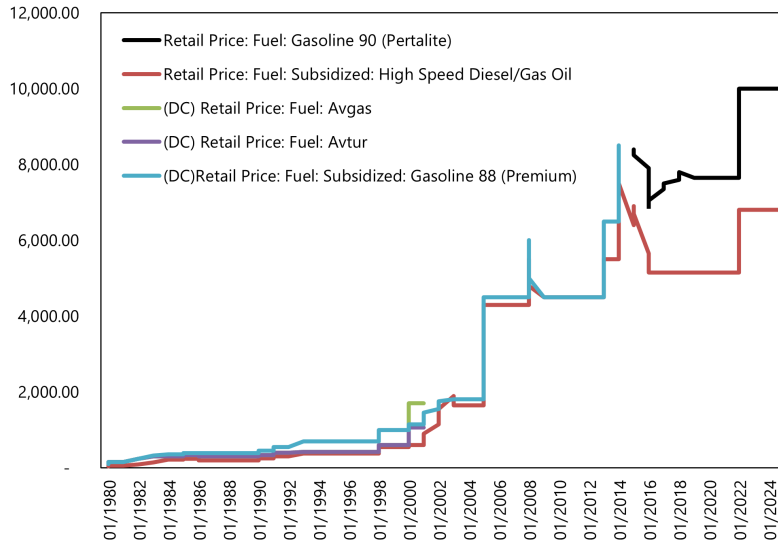


Figure 1: Retail Fuel Price in Indonesia, 1980-2025

Source: Statistics Indonesia (via CEIC).

Notes: Before 2001, the GoI provided subsidies for aviation turbine fuel (avtur) and gasoline (avgas), gasoline (RON 88), kerosene, and diesel. In 2002, the GoI removed avtur and avgas from subsidised petroleum. In 2015, gasoline RON 88 was replaced by RON 90.

Following prolonged period of constant subsidies, the subsidy regime underwent several reforms over the past two decades. After AFC in 1998, the GoI faced widening fiscal deficits, and this forced the GoI to cut spending. One of the major measures was to reform fuel subsidies. According to the Presidential Decree No. 45/2001, the GoI adjusted the following aspects related to petroleum subsidies: (i) The GoI was no longer

¹Before 2004, Indonesia was net-oil exporting countries. However, as the petroleum consumption grew exponential while the production was stagnant, Indonesia has become oil-importing countries (Dartanto, 2013, 2017).

providing subsidies for avtur and avgas; (ii) GoI increased retail prices to narrow the price gap; and (iii) the GoI gradually excluded the industry from accessing subsidised petroleum. The GoI also adjusted the retail price more frequently to match international oil price (Figure 1). At that time, the GoI essentially narrowed the objective of the subsidies to only protect households and small business from oil price volatility.

Until 2003, such measures raised protests then the GoI rolled back, once again adjusting fuel price less frequent (Yusuf et al., 2017; International Monetary Fund, 2013). As the global oil price kept rising, fuel subsidies were more than doubled from 2003 to 2004, threatening fiscal stability. This led the GoI to increase the retail price by around 100% (see Figure 1) and to completely revoked petroleum subsidies for industry.

After significant price adjustment in 2004, the GoI kept the price fixed for relatively long time. However, Indonesia was experiencing twin deficit and depreciated exchange rate in 2014. Between 2010 and 2014, total subsidies for petroleum was increased by 30.66% per year, rising from only IDR 82 Trillion to IDR 239 Trillion in 2014 (see Figure 2). Such a sharp increase was primarily driven by the surge of oil price during this period, compounded by the insistence of the government to keep the retail price unchanged. As a result, petroleum subsidies accounted for almost 15% of the total government expenditure, significantly widening the Indonesian fiscal deficits (Basri, 2017). In response, the GoI tried to manage the deficit by doubling retail price on subsidised petroleum (Yusuf et al., 2017; Basri, 2017).

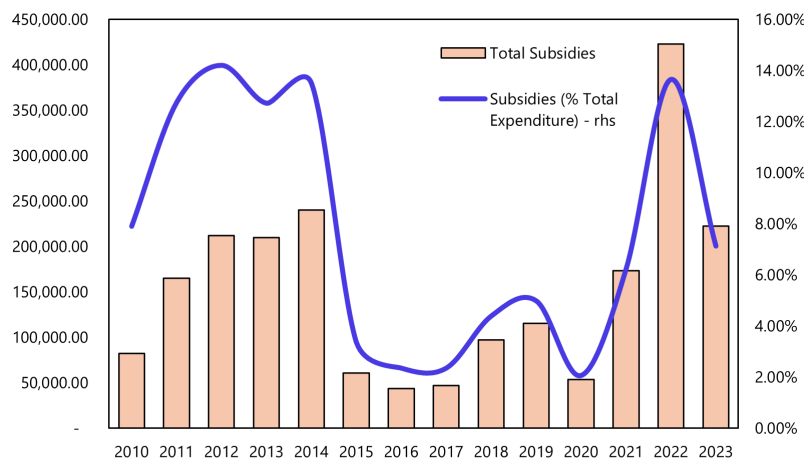


Figure 2: Government Expenditure on Petroleum Subsidies in Indonesia, 2010-2023

Source: The Audit Board of the Republic of Indonesia.

Notes: Total subsidies are represented in IDR billion. Total subsidies include implicit and explicit subsidies.

In 2015, the GoI continued a series of reform on petroleum subsidies. The first reform was removing gasoline RON 88 from subsidised petroleum, and the GoI introduced gasoline RON 90.² The initial objective was to adjust consumers' behaviour of using subsidised gasoline RON 88, replace it with higher RON gasoline until the consumers can shift to non-subsidised gasoline (RON 92). At the beginning, price of gasoline RON 90 was adjusted frequently, linked to the international oil price fluctuation. However this mechanism lasted for less than two years. In the mid 2017, the GoI has stalled this approach and again set the price of RON 90 fixed for a long time (Wardhana et al., 2024).

Such a reform reversal left Indonesia vulnerable to the subsequent shocks. When the oil price surged again since the beginning of 2021, the fiscal burden of fixed retail prices increased sharply. For petroleum subsidies, total (implicit and explicit) subsidies mounted to more than IDR 400 trillion, or equivalent to around 14 percent of the total state expenditure (see Figure 2). Consequently, the government was forced to adjust retail price of subsidised gasoline. However, inflation spiked from 1.87% (year-on-year) in 2021 to 5.51% (yoy) in 2022. To contain the adverse impacts, especially to the poor, the GoI provided unconditional cash transfers. Al Izzati et al. (2023) show that such measures can contain the negative effects.

Yet these recurrent episodes highlight a structural problem. The most immediate concern relates to the heavy fiscal burden created by petroleum subsidies. During the periods of rising global oil prices, the GoI can spend around 15% of the total budget only to provide petroleum subsidies. Such a substantial fiscal burden crowds out other priority public spending. Several studies highlight that the growing amount of fuel subsidies in Indonesia reduces fiscal space to promote economic growth, poverty alleviation, and clean energy transition (Dartanto, 2017; Attwood et al., 2022).

Beyond their fiscal implications, subsidies also attract criticism for its regressive distribution (Dartanto, 2013). Table 1 shows the distributive characteristics of gasoline consumption and subsidies across expenditure groups. The proportion of households that consume gasoline increases with the expenditure level. While nearly all of the

²According to the most recent regulation, the GoI provides subsidies for Peralite (RON90) and Bio-diesel. In this regard, there are several differences of mechanism between subsidies on Peralite (RON90) and Bio-diesel. For Peralite subsidy, the GoI sets the subsidized retail price and may adjust it with discretion any time. The GoI covers the price gap by compensating Pertamina, a state-owned enterprise mandated to distribute Peralite. This mechanism is referred as compensation or implicit subsidies. In contrast, the GoI provides a fixed per liter subsidies for Bio-diesel, i.e., IDR1,000 per liter, so that the subsidised retail price may change according to market price fluctuation. This is called explicit subsidies.

Table 1: Distributive Characteristics of Gasoline Consumption and Subsidies, 2023

Decile by Expenditure	Percentage of HH that Consume Gasoline	Average HH's Expenditure on Gasoline (IDR/HH/Month)	Average Volume of Gasoline Consumed (IDR/HH/Month)	Average Subsidies Received (IDR/HH/Month)
1	27.33%	104,909.90	9.72	22,541.30
2	50.45%	133,051.50	12.41	26,894.27
3	61.13%	153,609.70	14.10	29,606.15
4	68.78%	170,660.50	15.56	30,977.06
5	74.02%	190,886.40	17.12	32,150.62
6	78.30%	213,667.90	18.83	34,273.64
7	81.48%	247,181.10	21.45	36,703.37
8	85.39%	287,756.30	24.73	42,763.94
9	89.27%	362,026.50	30.55	50,139.50
10	91.11%	575,781.90	46.33	59,822.76

Source: Author's calculation from National Socio-economic Survey (March 2023).

Notes: Total subsidies received by the households are estimated using $Q_{gasoline} \times (P_E - P_G)$ where economic price P_E is calculated using the formula set by the government (see Decree of Minister of Energy and Mineral Resource Number 255.K/MG.01/MEM.M/2022) and P_G is obtained from the unit price of gasoline the consumer pays.

richest household consume gasoline, only about one-quarter of the poorest households do so. Among the gasoline consumers, the poorest households consume about 10 liters per month, significantly lower than the richest that can consume up to 46 liters. This pattern then implies the regressive nature of the subsidies as the richer households not only consume more gasoline but also receive larger nominal subsidies.

In response to those problems, many proposals of reform have been proposed. Some suggest that the GoI should completely phase out the subsidies but gradually and accompanied by some compensation to affected households (Dartanto, 2013; Ihsan et al., 2024; Wardhana et al., 2024). Such a reform could reduce fiscal burden, generate considerable fiscal savings that can be redirected to transfer to poor households, public investment and energy transition (Dartanto, 2013, 2017; Yusuf et al., 2017; Attwood et al., 2022; Fathurrahman et al., 2015). The other propose **targeted subsidies**. Instead of allowing all types of consumers to purchase subsidised petroleum products, under targeted subsidies scenario, only several types of consumers can access the subsidies for instance, the poorest 40 percent. Lastly, some also recommend that the GoI should change the pricing mechanism from ad hoc price adjustment to **dynamic pricing adjustment** (Wardhana et al., 2024).

3 Model

The model developed in this paper builds on Pieroni (2023) with some extensions. Following Pieroni (2023), the economy consists of households, union, final and intermediate good producers, central bank, and fiscal authority. Households are ex-ante homogeneous but ex-post heterogeneous due to idiosyncratic shocks and incomplete financial markets. Households supply labour through union, giving rise to nominal wage rigidity. Final good producers operate in perfectly competitive market and aggregate differentiated intermediate goods. Intermediate good producers compete in monopolistic market, use labour and petroleum as production inputs, and face Rotemberg-style price adjustment costs. Central bank sets nominal interest rates using Taylor rule. Fiscal authority collects consumption and labour income taxes and implements petroleum subsidies.

The key extensions of my model lies in how the market price of petroleum determined and the regime I consider in this model. Instead of considering the aggregate shocks from petroleum shortage, I consider the shocks come from price. In other words, petroleum prices are exogenously determined, reflecting the Indonesian economy as a price-taker in global oil market. This is consistent the fact that Indonesia is net-oil importing country. Furthermore, I consider five different petroleum regimes namely price subsidies for households with ad hoc price adjustments (baseline), no subsidies, automatic subsidised price adjustment, mean-tested transfers, and lump sum transfers.

3.1 Households

There is a continuum, mass number of infinitely lived households $h \in [0, 1]$ who consume two different commodities namely non-petroleum consumption $c_{f,t}$ and petroleum consumption $c_{e,t}$. The composite consumption c_t aggregates these two commodities using standard CES technology:

$$c_t = \left(\alpha^{\frac{1}{\sigma^h}} c_{f,t}^{\frac{\sigma^h}{\sigma^h-1}} + (1 - \alpha)^{\frac{1}{\sigma^h}} c_{e,t}^{\frac{\sigma^h}{\sigma^h-1}} \right)^{\frac{\sigma^h-1}{\sigma^h}} \quad (1)$$

where α represents the weighted between non-petroleum and petroleum consumptions and σ^h is the elasticity of substitution between $c_{f,t}$ and $c_{e,t}$.

Now let $p_{f,t}$ and $p_{e,t}$ as the price level for non-petroleum and petroleum consumptions. By the cost-minimizing of composite consumption in equation (1), we can define

the aggregate price p_t as the following $p_t = \left(\alpha p_{f,t}^{1-\sigma^h} + (1-\alpha) \bar{p}_{e,t}^{1-\sigma^h} \right)^{\frac{1}{1-\sigma^h}}$ and aggregate nominal consumption as $p_t c_t = p_{f,t} c_{f,t} + p_{e,t} c_{e,t}$ (see Appendix A.2.1 for details).

In this economy, households are maximizing their lifetime utility by choosing composite consumption c_t and real bonds b_t subject to real budget constraint, borrowing constraint and idiosyncratic shocks. Labour supply decision N_t is made not by the households but through unions. The heterogeneous households problem is given by.

$$V(e_t, b_{t-1}) = \max_{c_t, b_t} \frac{c_t^{1-\gamma}}{1-\gamma} - \frac{N_t^{1+\psi}}{1+\psi} + \beta \mathbb{E}_t V(e_{t+1}, b_t) \quad (2)$$

$$\text{s.t. } b_t + (1 + \tau^c) c_t = (1 + r_t) b_{t-1} + (1 - \tau^N) w_t e_t N_t + d_t + tr_t,$$

$$b_t \geq 0,$$

$$\log(e_{t+1}) = \rho_e \log(e_t) + \varepsilon_{e,t+1}, \quad \varepsilon_e \sim N(0, \sigma_e), \quad \mathbb{E}(e_t) = 1$$

where γ is elasticity of substitution, ψ is Frisch elasticity of labour supply, N_t denotes labour working hours, r_t is real interest rates, w_t is real wage, d_t is dividend in real term, tr_t is transfers from government, and τ^c and τ^N respectively represents distortionary consumption and labour income taxes.³ Lastly, e_t denotes idiosyncratic income shock such that it follows stationary AR(1) process with the cumulative distribution over states is defined by $\Gamma : (B, e) \rightarrow [0, 1]$.⁴

3.2 Market Structure

3.2.1 Wage Setting and Labour Supply

There is a continuum number of union $u \in [0, 1]$ aiming at maximize the overall households' economic benefit by choosing labour supply $N_t(u)$ and nominal wage level $W_t(u)$:

$$\max_{\{W_t(u), N_t(u)\}} \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \left[\int_0^1 \frac{W_{t+s}(u)}{p_{t+s}} - \frac{v'(N_{t+s}(u))}{u'(C_{t+s})} N_{t+s}(u) - \frac{\Psi^W}{2} \left(\frac{W_{t+s}(u)}{W_{t+s-1}(u)} - 1 \right)^2 N_{t+s} du \right]$$

³Under non transfer policy regimes, $tr_t = 0$.

⁴In addition to the real budget constraint defined above, it is derived from nominal terms

$$(1 + \tau^c) (p_{f,t} c_{f,t} + p_{e,t} c_{e,t}) + B_t = (1 + i_{t-1}) B_{t-1} + (1 - \tau^N) W_t e_t N_t$$

where B_t is nominal bond holding, W_t is nominal wage. To obtain budget constraint in real terms, we substitute $p_{f,t} c_{f,t} + p_{e,t} c_{e,t}$ with $p_t c_t$ and divide both sides with p_t

Subject to labour demand from competitive recruiting firms:

$$N_t(u) = \left(\frac{W_t(u)}{W_t} \right)^{-\theta^W} N_t$$

where θ^W is constant elasticity of substitution of labor, $C_t = \int g_{c,t}(b, e) d\Gamma_t$ is aggregate consumption, Ψ^W is coefficient of quadratic adjustment cost a la Rotemberg.

By symmetric equilibrium, the optimality condition of the labour supply results in following wage New Keynesian Phillips Curve:

$$\Psi^W \pi_t^W (1 + \pi_t^W) = \kappa^W \left(\frac{N_t^\psi}{C_t^{-\gamma}} - \mu^W w_t \right) + \beta \mathbb{E}_t \left[\pi_{t+1}^W (1 + \pi_{t+1}^W) \frac{N_{t+1}}{N_t} \right] \quad (3)$$

where $\kappa^W = \frac{\theta^W}{\Psi^W}$, $\mu^W = \frac{\theta^W - 1}{\theta^W}$, and π_t^W is wage inflation.

3.2.2 Final and Intermediate Goods Producers

There is a continuum competitive final goods producers aggregating differentiated intermediate goods $f \in [0, 1]$, with the following optimization problem:

$$\max_{Y_t, Y_t(f)} p_{f,t} Y_t - \int_0^1 p_t(f) Y_t(f) df \quad \text{s.t.} \quad Y_t = \left(\int_0^1 Y_t(f)^{\frac{\theta^F - 1}{\theta^F}} df \right)^{\frac{\theta^F}{\theta^F - 1}}$$

Given this problem, we can obtain demand for intermediate goods as follows

$$Y_t(f) = \left(\frac{P_t(f)}{p_{f,t}} \right)^{-\theta^F} Y_t$$

The production function is defined by

$$Y_t(f) = A_t N_t^{\alpha^F} O_t^{(1-\alpha^F)}$$

Intermediate goods producers maximize life-time profit by choosing inputs and price.

$$\max_{N_t(f), O_t(f), p_t(f)} \mathbb{E}_t \sum_{s=0}^{\infty} \Lambda_{t,t+s} p_{t+s} \left[\frac{p_{t+s}(f) Y_{t+s}(f)}{p_{t+s}} - w_{t+s} N_{t+s}(f) - \frac{p_{e,t+s}}{p_{t+s}} O_{t+s}(f) - \frac{\Psi^F}{2} \left(\frac{p_{t+s}(f)}{p_{t+s-1}(f)} - 1 \right)^2 Y_{t+s} \right]$$

Where Λ is stochastic discount factor and $p_{e,t}$ is the market price of petroleum.⁵. Recall that in Lagrangian form, the Lagrangian Multiplier can be interpreted as real

⁵In this model, we assume the subsidies are not provided to firms so firms face p_e not $\bar{p}_e$.

marginal cost $mc_t(f)$. Therefore, we can obtain the input demand as follows.

$$N_t(f) = \alpha_F \left(\frac{mc_t(f)}{w_t} \right) Y_t(f) \quad (4)$$

$$O_t(f) = (1 - \alpha_F) \left(\frac{p_t mc_t(f)}{p_{e,t}} \right) Y_t(f) \quad (5)$$

Lastly, by imposing symmetric equilibrium, we can define the following inflation NKPC:

$$\pi_t^F (1 + \pi_t^F) = \kappa^F (mc_t - \mu^F \frac{p_{f,t}}{p_t}) + \Lambda_t \mathbb{E}_t \left[\pi_{t+1}^F (1 + \pi_{t+1}^F) (1 + \pi_{t+1}) \frac{Y_{t+1}}{Y_t} \right] \quad (6)$$

where $\kappa^F = \frac{\theta^F}{\Psi^F}$, $\mu^F = \frac{\theta^F - 1}{\theta^F}$, π_t^F is inflation for intermediate goods. Lastly, it is also worth noting that $\Lambda_t = \frac{1}{1+r_{t+1}}$.

3.2.3 Oil Market

Indonesia is a price taker in oil market as it has become net-oil importing country in 2004. Given this fact, we consider that the market price of oil is exogenous following AR(1) process such as in Omotosho (2022).

$$\log(p_{e,t}) = \rho \log(p_{e,t-1}) + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2) \quad (7)$$

where $\rho_e \in (0, 1)$ and $\mathbb{E}(p_e) = 1$.

3.3 Monetary Policy

Central bank follows Taylor-type rule by choosing nominal interest rate

$$i_t = r_{ss} + \phi_\pi \pi_t \quad (8)$$

where r_{ss} is the steady state level of real interest rate and we assume that the monetary policy is passive by only responding to inflation, $\phi_\pi > 1$.

Fisher equation links nominal to real interest rate

$$1 + r_t = \frac{1 + i_{t-1}}{1 + \pi_t} \quad (9)$$

3.4 Fiscal Policy

The government budget constraint in nominal terms is defined as

$$\tau_t^N w_t N_t + \tau_t^c c_t + b_t^g = (1 + r_t) b_{t-1}^g + G_t + tr_t + s_t \quad (10)$$

where b_t^g is total bonds issued, b_{t-1}^g is matured bonds, G_t is government consumption, tr_t is transfer, and s_t is spending on petroleum subsidies.⁶

Total government spending on petroleum subsidies is thus given by

$$s_t = \frac{\bar{p}_{e,t} - p_{e,t}}{p_t} \left(\int c_{e,t} d\Gamma_t \right)$$

where $\bar{p}_{e,t}$ and $p_{e,t}$ be subsidised and non-subsidised petroleum prices such that $\bar{p}_{e,t} \leq p_{e,t}$, $c_{e,h,t}$ denotes petroleum consumption by household h , and Γ_t is the cumulative distribution function of the households.

To ensure that the equilibrium existence of government budget constraint, we assume that the fiscal policy is active by adjusting labour income tax rate according to the following rule:

$$\tau_t^N = \tau_{ss}^N + \iota_b (b_{t-1}^g - b_{ss}^g) \quad (11)$$

where τ_{ss}^N and b_{ss}^g are steady state level of income tax rate and real bonds. $\iota_b \in (0, 1)$ is sensitivity coefficient.

Furthermore, the fiscal authority operates through two broad policy approaches: price controls and cash transfers. Under the price control approach, the model considers three regimes: *ad hoc* price adjustment (baseline regime), automatic price adjustment, and no subsidy regime. Under the cash transfer approach, the government may instead provide either mean-tested transfers or universal lump sum transfers.

3.4.1 Price Control Regime: Ad Hoc Price Adjustment, Automatic Price Adjustment, and No Subsidies

Under price control regime, the government set the subsidised price of petroleum according to the following general rule:

$$\bar{p}_{e,t} = \omega \bar{p}_{e,t-1} + (1 - \omega) \varphi p_{e,t} \quad (12)$$

⁶Under non transfer policy regimes, $tr_t = 0$.

where $\omega \in [0, 1)$ regulates the persistence of the changes of subsidised price and $\varphi \in (0, 1]$ ensures that in the long run, the government is still providing subsidies. Additionally, $tr_t = 0$ under this regime.

Price control rule in (12) follows Ginn (2024). This rule can be used to define three kind of policies (i.e., Ad Hoc Price Adjustment, Automatic Price Adjustment, and No Subsidies) by varying ω and φ .

Under the **ad hoc price adjustment arrangement (hereafter baseline regime)**, the government adjusts the subsidised price of petroleum less frequently and still provides the subsidies in the long run. To that end, the government sets $\omega \in (0, 1)$ and $\varphi \in (0, 1)$. For our baseline analysis to mimic Indonesian economy, we set $\omega = 0.9$ and $\varphi = 0.6$. This suggests that the subsidised price of petroleum set by the government will be persistent in response to shocks and 40% lower than the market price (Yusuf et al., 2017).

For **automatic price adjustment policy**, we set $\omega \in (0, 1)$ and $\varphi = 1$. This suggests that the government will provide no subsidies in the long run but still manage to constrained the fluctuation of subsidised price caused by market price shocks. Lastly, the government can also provides **no subsidies** at all by choosing $\omega = 0$ and $\varphi = 1$. This simply means that the equivalence of market and subsidised price, $\bar{p}_{e,t} = p_{e,t}$

3.4.2 Cash Transfers Regime

Under transfer regimes, the government replaces fuel subsidies with transfers that increase with oil prices. Equation (13) provides a unified framework that nests the two transfer regimes.

$$tr_{h,t} = \left(\frac{\varphi^{tr}}{(1 + \kappa^{tr} \exp(y_{h,t} - \bar{y}))} \right) \frac{p_{e,t}}{p_t} \int c_{e,t} d\Gamma_t \quad (13)$$

where $\varphi^{tr} \in (0, 1)$, $\kappa^{tr} \geq 0$, and $\bar{y}_t$ is the pre-transfer income cutoff. More specifically, φ^{tr} and κ^{tr} respectively regulate the maximum transfers derived the portion of aggregate petroleum consumption and the slope of the amount of transfer across income level.

Deriving from the above rule, we have two policies we can define. First, the government conducts **mean-tested transfers** by setting up $\varphi^{tr} \in (0, 1)$ and $\kappa^{tr} > 0$. Under this policy, there are two crucial features to be exhibited. Following Auclert et al. (2021), the first feature is the exposure over the petroleum price dynamics. When the petroleum price shocks exist, the government provides more transfer to the households. Second, the transfer varies according to the level of income in a sigmoid manner. The

higher the income the less the transfers. Furthermore, the government can also conduct **universal cash transfers** (i.e., everyone gets the same amount of transfers) if it sets $\varphi^{tr} \in (0, 1)$ and $\kappa^{tr} = 0$.

3.5 Equilibrium

The competitive equilibrium is defined as sequence of (a) distributions over households states $\{\Gamma_t\}_{t=0}^\infty$; (b) households' value function $\{V_t, g_{c,t}, g_{b,t}\}_{t=0}^\infty$; and (c) aggregate quantities and prices $\{C_t, C_{e,t}, C_{f,t}, Y_t, N_t, O_t, b_t^g, \tau^N, G_t, s_t, d_t, tr_t, i_t, r_t, w_t, p_t, p_{f,t}, \bar{p}_{e,t}, \pi_t, \pi_t^W, \pi_t^F\}_{t=0}^\infty$ such that households' optimization holds, firms' optimization holds, government budget constraint holds, Taylor rule holds, and market clears (see Appendix A.3 for the system of equations).

3.6 Calibration

Parameters in this model are calibrated to closely match several key distributional and petroleum-related characteristics of the Indonesian economy: **(a) Wealth Gini coefficient; (b) Income Gini Coefficient; (c) Subsidy expenditure to government consumption Ratio; and (d) Share of petroleum consumption.** Table (2) reports the calibrated parameter values used in the model. I group the parameters into four blocks: households, union and firms, policy, and oil price shocks.

- **Households**

Parameter α regulates the fraction of households' non-petroleum consumption. I assign $\alpha = 0.965$ to match the observed share of petroleum consumption c_f in Indonesia. For the elasticity substitution between petroleum and non-petroleum consumption (σ^h), I follow Gagliardone and Gertler (2023) who argue that the consumption between these two commodities are highly complementary, implying $\sigma^h < 1$. Furthermore, I follow the Auclert et al. (2021) for the Frisch elasticity of labour supply (ψ), CRRA (γ). Lastly, idiosyncratic income shocks follow an AR(1) process with persistence $\rho_e = 0.96$ and the innovations of the shocks $\sigma_e = 0.2$, discretised using Rouwenhorst Discretisations with five states $n_e = 5$.

- **Union and Producers**

For the union and firms block, I adopt standard values used in literature for the constant elasticity of labour and the adjustment costs. However, I set the price

Table 2: Model Parameters

Parameters	Description	Value	Source
<i>Households</i>			
β	Discount factors	0.938	Annualized from Auclert et al. (2021)
α	Distribution parameter of non-petroleum consumption	0.965	To match petrol consumption share
σ^h	Elasticity of Substitution	0.5	Gagliardone and Gertler (2023)
ψ	Frisch elasticity of labor supply	1	Auclert et al. (2021)
γ	Coefficient of relative risk averse	2	Pieroni (2023); Auclert et al. (2021)
τ^c	Consumption tax	0.11	VAT rate in Indonesia
ρ_e	Idiosyncratic shocks persistence	0.96	To match Indonesian Gini
σ_e	Innovation of idiosyncratic shocks	0.2	To match Indonesian Gini
n_e	Number of productivity states	5	Author
<i>Union and Firms</i>			
θ^W	Constant elasticity of substitution of labor	10	Pieroni (2023)
Ψ^W	Cost of wage adjustments	75	Author
θ^F	Constant elasticity of substitution of input	10	Pieroni (2023)
Ψ^F	Cost of price adjustments	75	Author
<i>Policy</i>			
φ^{tr}	Intensity of transfers (mean-tested)	0.705	Target: $tr = 0.0198$
	Intensity of transfers (lump sum)	0.613	Target: $tr = 0.0198$
κ^{tr}	Slope of mean-tested transfers	10	Author
φ_π	Policy rate responses to inflation	1.25	Passive Monetary Policy
<i>Petroleum Price</i>			
ρ	Petroleum price persistence	0.86	Author
σ_ϵ	Innovation of petroleum price	0.48	Author

adjustment cost (Ψ^F, Ψ^W) lower than Pieroni (2023) who set it at 100. This choice reflects the empirical evidence that price and wage rigidities tend to be lower in developing economies, where adjustment may occur more frequently (Choudhary et al., 2016; Klenow and Malin, 2010).

• Policies

I calibrate $\varphi^{tr} = 0.705$ for mean-tested transfers and $\varphi^{tr} = 0.613$ for lump sum transfers to match the aggregate transfers with total of subsidies under baseline scenario $tr = 0.0198$. This rationale for this is to capture the effect of reforming the policy from subsidies to mean-tested or lump sum transfers with the same amount of budget allocation. Lastly, I set the monetary policy to be passive ($\varphi_\pi > 1$) and fiscal policy to be active through income tax rate ($\iota_\tau = 0.5$). While there is no consensus on the value of how responsive the income tax rate is for Indonesian economy Juhro et al. (2022), I set ι_τ so that b^g is converged in the long run.

• Oil Price Shocks

I estimated AR(1) process using annual WTI crude oil price. This results in

Table 3: Key Variables Comparison

	Model	Data
Gini Coefficient (Assets)	0.7326	0.7454
Gini Coefficient (Income)	0.3449	0.3674
Subsidies to Government Consumption	0.0765	0.0768
Share of Petroleum Consumption	0.0447	0.0443

Notes: The variables generated by the model is steady state value under baseline regime (price subsidies) while those generated by the data is the mean over certain periods. For Asset Gini Coefficient, the data is obtained from World Inequality Data (WID) from 2000-2023. Income Gini Coefficient is obtained from World Income Inequality Database (WIID). Ratio of subsidies to government consumption is collected from CEIC and the Audit Board of Indonesia throughout 2010-2023. Lastly, share of petroleum consumption obtained from Living Cost Survey 2022.

persistence parameters $\rho = 0.86$ and $\sigma_\epsilon = 0.48$. For the simulation, I assume one percent deviation of oil prices.

Table (3) shows key variables comparison between the model and real world data. Based on the model's structure and the calibrated parameters, variables generated by the model are very close to real world data. For both wealth-based and income-based Gini coefficient the model generates are 0.73 and 0.34 which are reasonably close to the actual data, 0.74 and 0.36. Furthermore, the amount of subsidies and the share of petroleum consumptions simulated from the model are similar to the real world data. This suggests that our model is able to capture the core characteristics of Indonesian economy.

4 Results and Discussions

This section is divided into two main parts. The first discusses the policy reform analysis, focusing on both long run outcomes of each policy regime and the transitional dynamics associated with reforms away from the baseline regime. The second part evaluates the effectiveness of petroleum subsidies as a stabilisation instrument in response to oil price shocks.

4.1 Policy Reform Analysis

4.1.1 Long Run Outcomes

Figure (3) exhibits the relationship between the fraction of petroleum price subsidies φ and the long run economic outcomes. The results suggest that although reducing subsidies raises price level, it improves broader economic performance through lower fiscal distortions and more efficient resource allocation. In particular, lower subsidy intensity stimulates output, improves aggregate welfare, reduces inequality, and strengthens the government's primary balance.

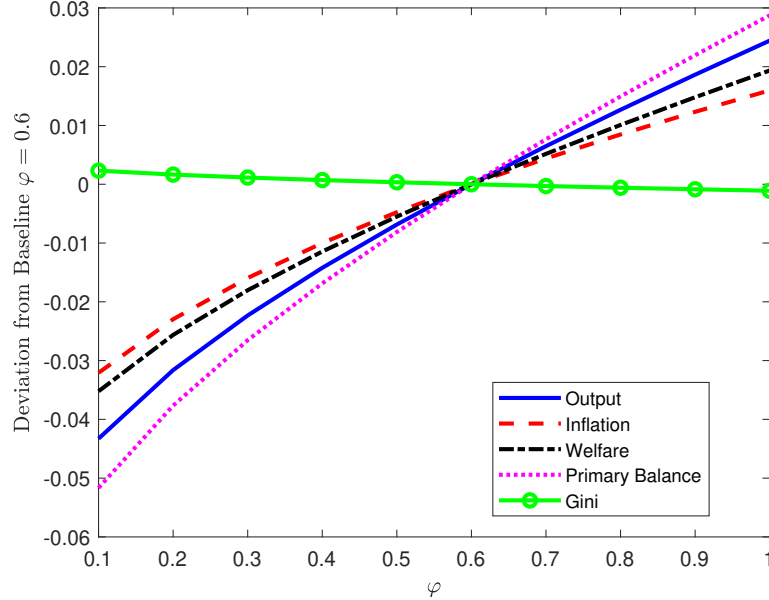


Figure 3: Impacts of φ on Macroeconomy, Primary Balance, Distribution, and Welfare

Notes: This figure exhibits the percentage difference between baseline portion of subsidies with $\varphi = 0.6$ to other portions. Higher φ corresponds to lower portions of subsidies paid by the government. In this Figure, the government does not provide transfers.

Reducing subsidy intensity, which corresponds to higher φ raises price level yet improves aggregate output. Relative to baseline scenario, if the government covers 90 percent of the petroleum unit price $\varphi = 0.1$, price level is 3 percent lower but output is contracted by more than 4 percent. In contrast, eliminating subsidies improves the output by more than 2 percent although the price level becomes 1.6 percent higher than the baseline regime. Furthermore, I also find that inflationary cost of subsidy reduction diminishes relative to its output gains as the subsidy intensity declines. Specifically, the gap between changes in output and inflation turns from negative to positive as φ increases.⁷ These results indicate that petroleum subsidies generate distortions, specifically through their effects on relative prices and input choice allocation (Yusuf et al., 2017). Subsidies distort real marginal costs and inputs choice from the efficient level and thus suppress the output. This result is consistent with Plante (2014) who argues that subsidies create significant distortions to the economy which adversely impact the GDP.

Furthermore, our model shows that wealth inequality declines and welfare improves in the long run as φ increases. The underlying transmission is straightforward. Price subsidy is inherently regressive as richer households tend to consume more petroleum. Reducing subsidies (lower φ) will disproportionately reduce the benefit to the rich households. Moreover, more efficient allocation due to subsidies removal increases labour productivity which in turn raises real wages, income, consumption, and thus the aggregate welfare. This aligns with Dennis (2016) who shows that, for the case of Indonesia, households' welfare will be improved as the petroleum subsidies revoked.

Lastly, our model shows that decreases in subsidies (higher φ) improve primary balance. This result is straight forward, and there are at least two transmissions explaining how. First, reduction in subsidies unambiguously reduces the total spending, leading to an immediate improvement in the fiscal position. Second, more efficient allocations due to lower subsidies raise real wages and consumption which in turn increase tax revenue.

Although subsidy removal entails an output-inflation tradeoffs, the long run consequences of subsidy removal are more favourable by any means. In this part, I extend the analyses by incorporating transfers from government. More specifically, I consider two transfer schemes: targeted transfers and (universal) lump sum transfers.

⁷The gap here is defined as the log difference between baseline and comparison: $gap_i = \log\left(\frac{y_i}{p_i}\right) - \log\left(\frac{y_{base}}{p_{base}}\right)$ where $i \in \{0.1, 0.2, \dots, 0.9\}$. This measurement captures the severity of output-inflation tradeoffs compared to the baseline policy.

Table 4: Steady State Comparison

Variables	No Subsidies	Mean-tested Transfers	Lump Sum Transfers
Welfare	0.0194	0.0264	0.0251
Y	0.0244	0.0130	0.0149
p	0.0160	0.0160	0.0160
c	0.0230	-0.0005	0.0028
c_f	0.0312	0.0075	0.0107
c_e	-0.2013	-0.2196	-0.2171
Primary Balance	0.0287	0.2101	0.1358
B^g	0.0452	0.0444	0.0438
τ^N	0.0000	0.1756	0.1540
Policy Rate	0.0000	0.1771	0.1056
N	-0.0085	-0.0196	-0.0178
O	-0.1916	-0.2005	-0.1991
w	0.0333	0.0333	0.0333
Profit	0.0083	-0.0029	-0.0011

Notes: Numbers in this table are representing the percentage difference from the baseline scenario. The steady state results for no subsidies and automatic price adjustment regimes are identical.

Table (4) reports the long run outcomes comparison between baseline and alternative regimes. The first key result is that the baseline scenario delivers the lowest welfare among all regimes. Mean-tested/targeted transfers regime produces highest welfare improvements by 2.64 percent compared to baseline. This is followed by lump sum transfers and no subsidies which improve the long run welfare by 2.51 percent and 1.94 percent compared to the baseline, respectively. This result is consistent with Ginn (2024) for the case of Malaysian economy in which they find that the welfare outcomes of targeted transfers outperform other regimes such as price control and uniform transfers.

Although mean-tested transfers policy delivers the largest welfare improvements compared to other regimes, providing no subsidies and transfers yield the highest level of output. Specifically, mean-tested and lump sum transfers only boost the output by 1.3 percent and 1.5 percent respectively, while no subsidies regime increase the output by 2.4 percent compared to the baseline. This result suggests that although transfers improve welfare, they still have a distortion to the economy. In the baseline scenario, price subsidies distort relative prices, leading to inefficient input choices and resource misallocation. Subsidy removal eliminates this distortion. Under transfer regimes, although firms face undistorted input prices and can allocate inputs efficiently, transfers introduce distortions on the household side. On the one hand, transfers disincentivise

supply labour, which dampens output relative to the no-subsidy case (i.e., income effects). On the other hand, transfers require increases labour tax income for financing. This raises substitution effects for labour supply. As a result, transfers mitigate the distortions from price controls but do not fully eliminate inefficiencies in the economy.

Furthermore, fiscal positions improve under alternative regimes. The primary balance increases by 21.01 percent under mean-tested transfers regime, 13.58 percent under lump sum transfers regime, and 2.87 percent under no subsidies. Somewhat counterintuitive, public debt in alternative regime increases in the long run. More specifically, it increases by 4.5 percent in no subsidies regime and 4.4 percent in mean-tested and lump sum transfers regimes. However, this simply reflects the government's stronger fiscal ability to service a higher level of debt.⁸ Moreover, transfers-based regimes should manage higher primary balance to finance higher debt services due to higher interest rates. This indicates the distortion raised by transfers: Transfers disincentivise savings, especially low-asset households so that the equilibrium of interest rates rises.

Table 5: Distribution Coefficients

Variables	Baseline	No Subsidies	Mean-tested Transfers	Lump Sum Transfers
Assets Gini Coefficient	0.7347	0.7338	0.7333	0.7346
Assets Share of Top 10 Percent	0.5383	0.5373	0.5411	0.5368
Assets Share of Bottom 50 Percent	0.0148	0.0151	0.0160	0.0144
Assets Share Ratio	36.3483	35.4785	33.7794	37.2009
Income Gini Coefficient	0.3451	0.3457	0.3396	0.3407
Income Share of Top 10 Percent	0.2573	0.2575	0.2554	0.2559
Income Share of Bottom 50 Percent	0.2682	0.2679	0.2717	0.2711
Income Share Ratio	0.9592	0.9615	0.9403	0.9440

Notes: The steady state results from no subsidies and automatic price adjustment regime are identical.

Table (5) exhibits the distributional outcomes in terms of wealth and income. The analyses consider not only Gini coefficients but also the assets and income share held by the top 10 percent and bottom 50 percent. In terms of wealth inequality, I find

⁸In the long run, intertemporal budget constraint is defined as follows:

$$\underbrace{T_{ss} - G_{ss} - tr_{ss} - s_{ss}}_{\text{Primary Balance}} = r_{ss} b_{ss}^g$$

This long run budget constraint simply states that higher primary balance increases the ability to service a higher level of debt. Note that the interpretation for the short run is different.

that alternative regimes improve equality yet limited. In contrast, income inequality declines more substantially under mean-tested transfers. Income-based Gini coefficient decreases by 0.5 basis points, and the income share of bottom 50 percent increase by 0.01 percentage points. Such an improvement reflects the progressive nature of mean-tested transfers under which higher-income households receive relatively lower transfers than low-income households.

4.1.2 Transitional Dynamics of Policy Reform

Alternative policy regimes provide long run economic gains. However, how do policy transitions unfold? In particular, is the transition smooth, or does it entail substantial short run adjustment costs?

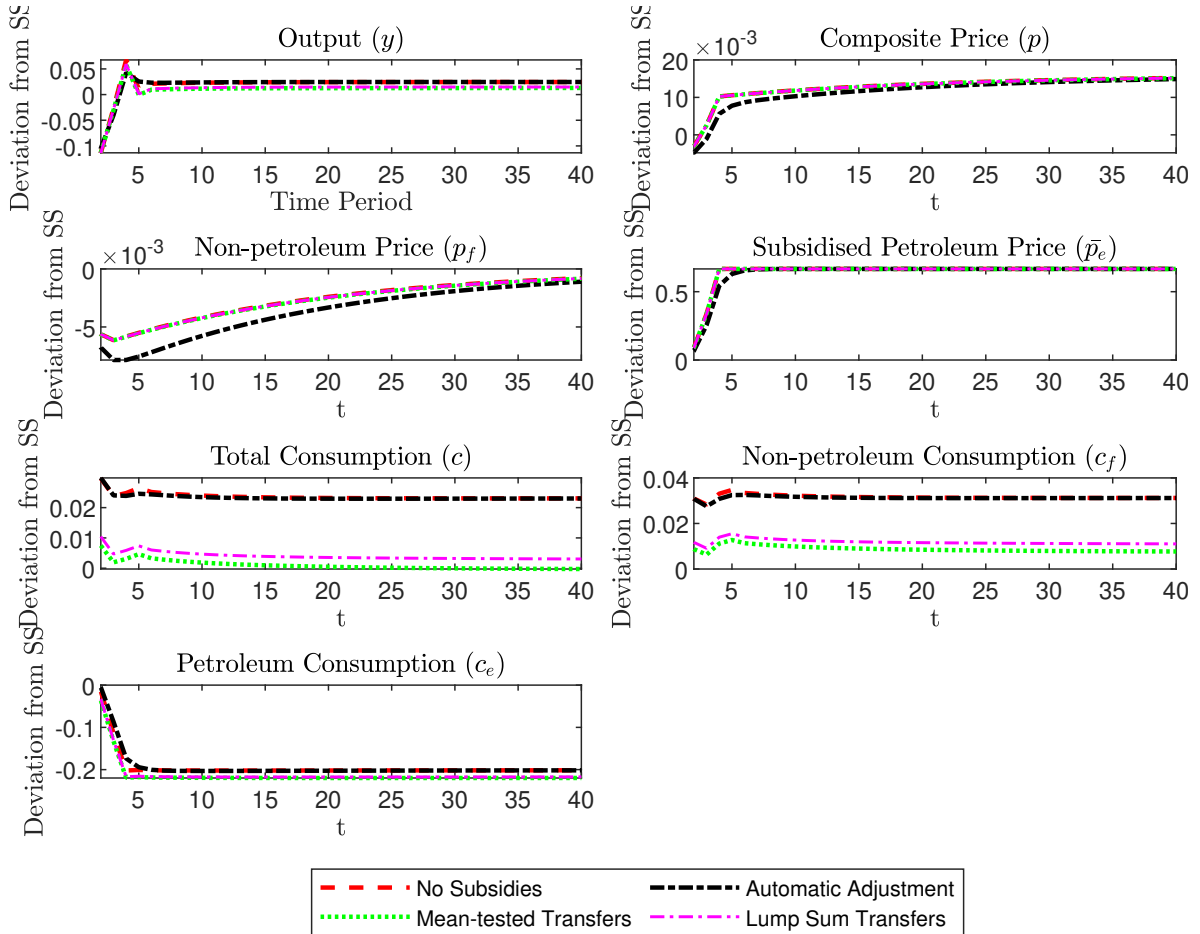


Figure 4: Policy Transitions I

Figures (4) to (6) present the transitional dynamics of policy reforms from the base-

line regime to alternative regimes. I consider gradual transitions where the government phases out the subsidies over four periods. This specification accommodates the idea of gradual removal as in Dartanto (2013); Ihsan et al. (2024); Wardhana et al. (2024); Yusuf et al. (2017); Attwood et al. (2022); Fathurrahman et al. (2015).

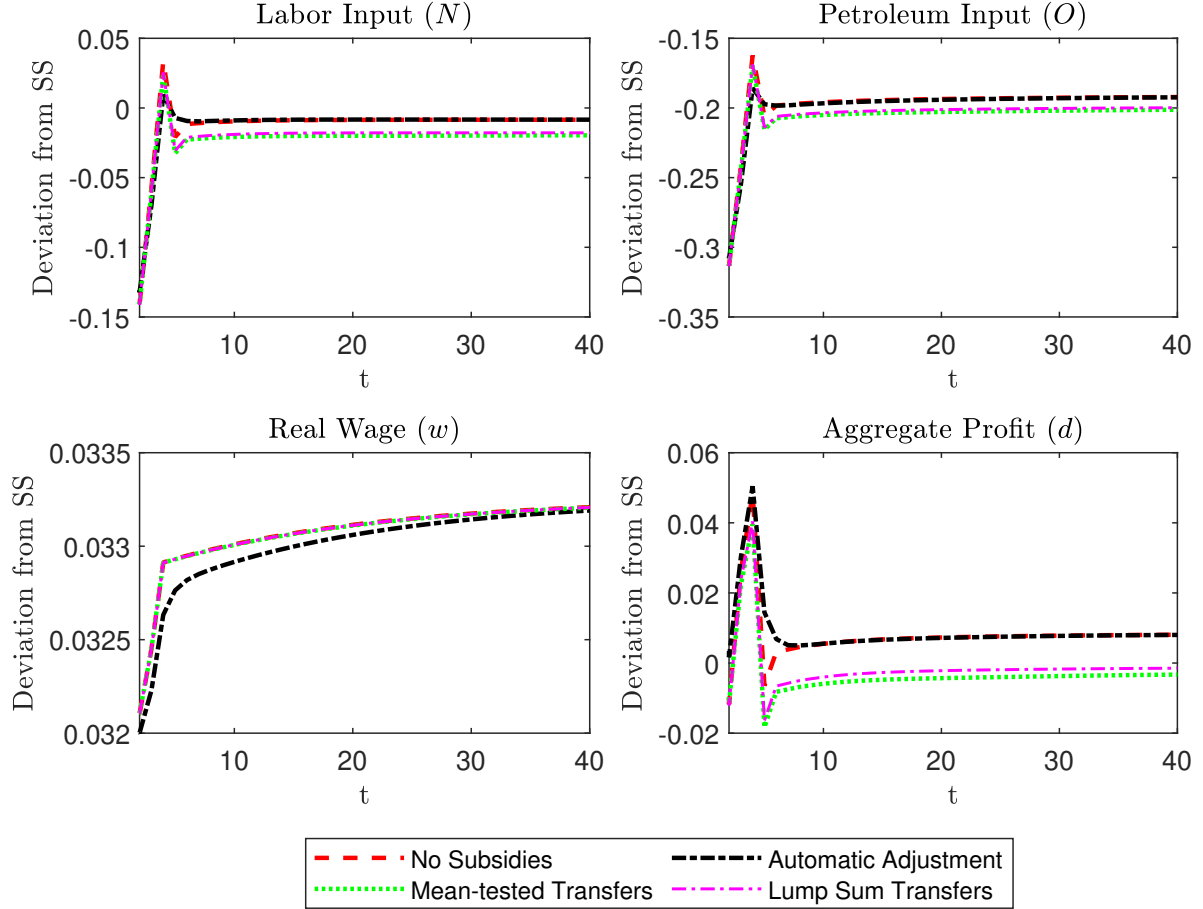


Figure 5: Policy Transitions II

In general, I find similar patterns across all reform scenarios. During the initial periods of reform, output contracts significantly while prices gradually increase, indicating that reforms may act as supply shocks to the economy. The initial output contraction is primarily driven by declines in labour and petroleum input. Specifically, labour input decreases by more than 10 percent while petroleum input reduces by around 30 percent. During the initial reform, firms recalibrate the input choices before it converges to new long run equilibrium, causing initial output contraction. This result is consistent with Yusuf et al. (2017) who find that subsidy removal causes the decline in employment in the early phase of the reform. In addition, real wage gradually increases toward the

new steady state, implying improvements in productivity.

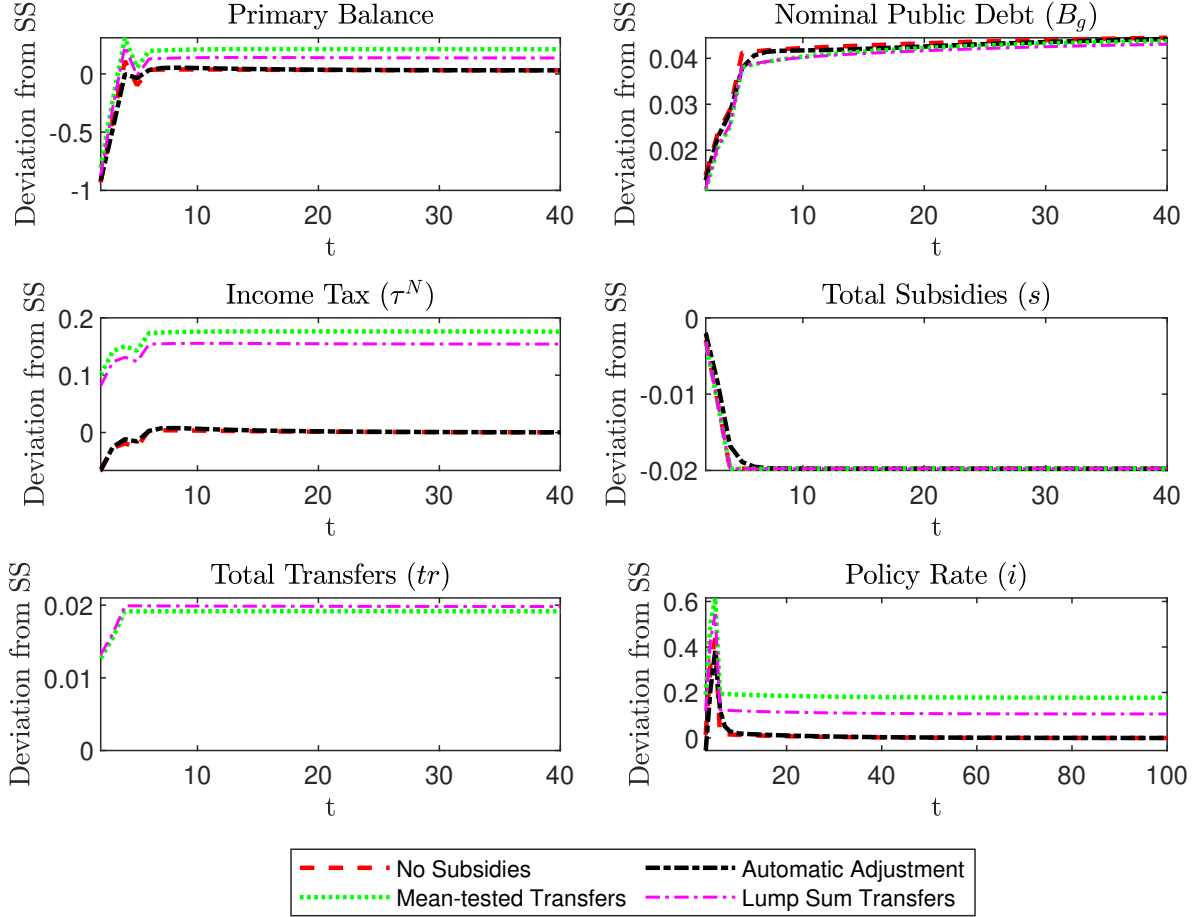


Figure 6: Policy Transitions III

Furthermore, general price level increases gradually in responses to the reform, primarily caused by the petroleum price. Petroleum price increases by 60 percent as government phases out the subsidies. In contrast, the price of non-petroleum goods initially decreases following the reform before gradually returning to the normal level. Changes in relative prices lead a decline in petroleum consumption and an increase in non-petroleum consumption. Under no subsidies and automatic price adjustment regimes, non-petroleum consumption increases by 2 percent whereas it rises only less than 1 percent under transfers-based regimes.

Finally, Figure (6) presents the fiscal dynamics in response to reforms. The primary balance drops significantly during the initial periods of reform, primarily due to reductions in tax revenue following contractions in the labour market. This in turn causes the surges in public debt. However, as the primary balance converges to the new steady

state level, public debt stabilises at the new long run equilibrium. Notably, tax rates increase significantly under transfers-based regimes. This is because the government requires to provide the transfers during the transition period.

Results presented in this section imply that reforms are not smooth, as they are associated with short-run economic contractions. Despite the long run and short run outcomes under alternative regimes outperform the baseline regime, unpleasant short run consequences of a reform may rationalise why is government reluctant to implement reforms. Alleyne et al. (2013) illustrate that the barriers to reform are not only related to political aspects, such as lack of confidence in the government, but also economic aspect including potential harmful impact on the poor and general economic conditions.

4.2 Short Run Dynamics

4.2.1 Impacts of Price Control Persistence ω and Portion of Subsidies φ

This section examines how the persistence parameter ω and share of government's unit price burden φ in price subsidy rule affect the cumulative responses of macroeconomic dynamics and discounted welfare following oil price shocks. Specifically, higher (lower) ω suggests more (less) persistent changes of subsidised petroleum price in response to oil price shocks, while lower (higher) φ indicates that the government carries more (less) fraction of a unit price subsidy. Figure (7) reports the cumulative effects of output, price, labour inputs, and discounted welfare in response to oil price shocks.

The cumulative effect of output is exhibited in Panel (a). Since oil price shocks act as negative supply shocks, a higher cumulative response corresponds to a smaller contraction in output relative to the baseline. In general, higher persistence ($\omega \rightarrow 1$) and larger government's share ($\varphi \rightarrow 0$) amplify the negative effects of oil price shocks on output fluctuations. The highest cumulative effect of output thus happens when $\omega = 0$ and $\varphi = 1$ which corresponds to no subsidies regime.

Panel (b) displays the cumulative effect of price level following oil price shocks. As the shock exerts upward pressure on the composite price level, a larger cumulative response reflects greater price fluctuations. In this regard, I find that price fluctuations are cushioned when the government lowers persistence ($\omega \rightarrow 0$) and provides more subsidies ($\varphi \rightarrow 0$). Allowing petroleum price fluctuates as the market price (i.e., no subsidies) results in relatively higher price fluctuations, although still moderate in magnitude. Moreover, I find that automatic price adjustment regime (i.e., $\omega \in (0, 1)$ and $\varphi = 1$) generates larger fluctuations than no subsidy regime, indicating relatively

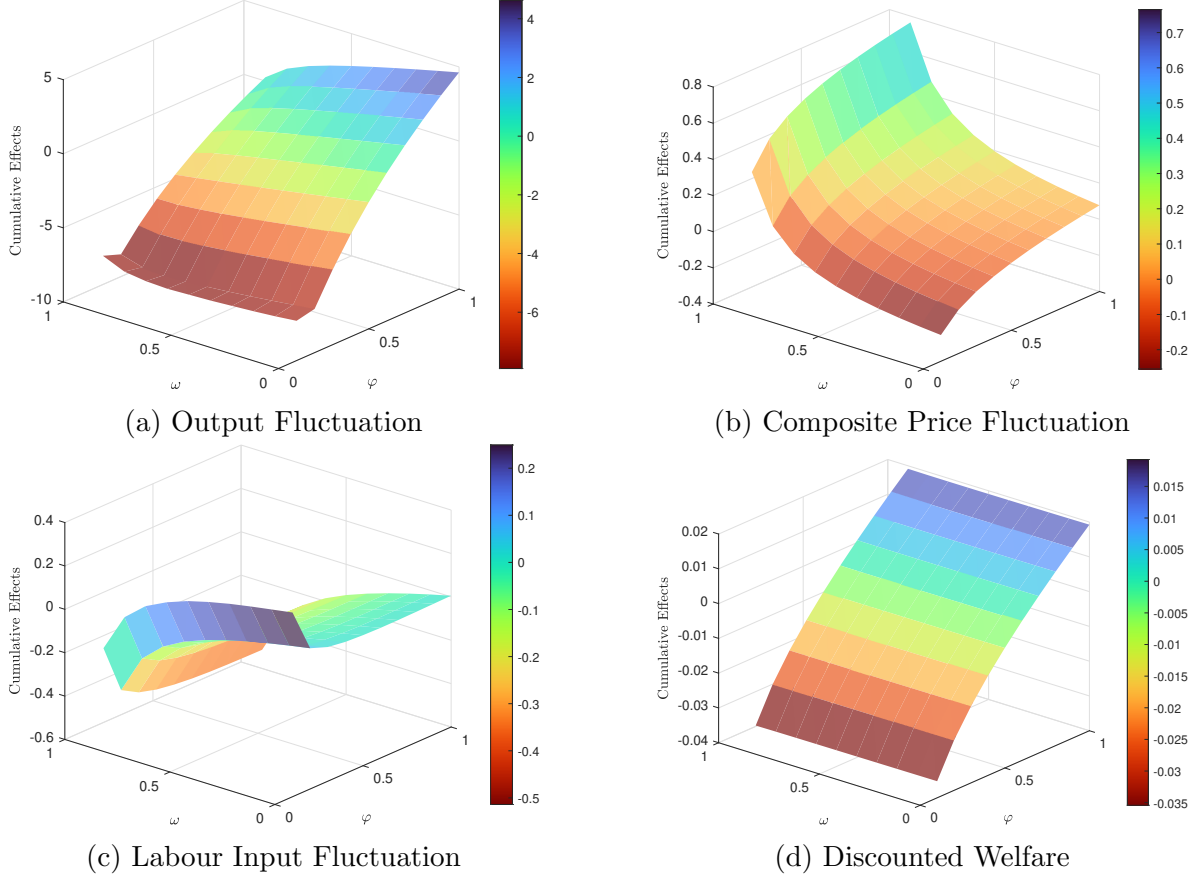


Figure 7: Cumulative Responses to Oil Price Shocks

Notes: This figure exhibits the percentage difference from baseline policy with $\phi = 0.6$ and $\omega = 0.9$. Higher ϕ indicates lower portions of subsidies paid by the government while higher ω suggests higher persistence. To calculate the cumulative effect of macroeconomic variables, I use this formula: $x(\omega, \phi) = \sum_{t=0}^T x_t(\omega, \phi; p_{e,t}) / x_{ss}(\omega, \phi) - 1$ where $x \in \{y, p, N\}$ and $x_t(p_{e,t})$ is the variable outcome as function of petroleum market price fluctuation due to shocks, x_{ss} is the variable level at steady state, and $T = 100$. For the discounted welfare, I use this formula: $W(\omega, \phi) = \sum_{t=0}^T \beta^{t-1} \left(\sum_{j=1}^3 \Gamma(j) V_t(j, b, \omega, \phi) \right) / T$. The government does not provide transfers, $TR = 0$.

weaker price stabilisation properties.

Oil price shocks bring downward pressure on labour inputs so that the lower cumulative response of labour indicates larger contractions. As presented in Panel (c), I find that higher persistence ($\omega \rightarrow 1$) leads to larger contractions in labour input compared to the baseline regime. Consistent the results of price fluctuations, I find that automatic price adjustment regime ($\omega \in (0, 1)$ and $\varphi = 1$) generates larger labour contractions than no subsidy regime, suggesting worse labour outcomes. Lastly, Panel (d) reports the cumulative discounted welfare following oil price shocks. I find that reducing unit price subsidies ($\varphi \rightarrow 1$) leads to substantial improvement in the discounted welfare. In contrast, persistence parameter ω only has limited effect on welfare.

The results in this section highlight several key insights. First, there is a clear tradeoff between price and output fluctuations. While higher persistence ω and larger subsidy share φ help cushion the price fluctuations, they exacerbate the contractions in output and labour. However, eliminating subsidies and allowing price to fluctuate according to the market result in substantial smaller output fluctuations but only moderate price volatility. This indicates that although the tradeoff persists, it becomes less pronounced under no subsidies. Furthermore, reducing subsidies also improves the discounted welfare throughout the period.

4.2.2 Macroeconomic Stability and Fiscal Dynamics

This section extends the analysis by considering not only price control regime but also transfers regimes. Specifically, I focus on how these policies can maintain macroeconomic stability and fiscal sustainability. Figures (8), (9), and (10) report the impulse response following a one-percent standard deviation shocks in oil price.

Oil price shocks act as supply shocks, it reduces output and escalates price level (see Figure 8). However, there are clear distinctions across policy regimes. Under baseline regime, output shrinks by 2.2 percent from the steady state in the first period, while price level increases only by 0.06 percent. Under automatic price adjustments regime, output contraction is smaller, but inflation is higher than baseline regime. Specifically, output declines by around one percent, but price rises by around 0.2 percent. Furthermore, under no subsidies and transfer-based regimes, output declines even smaller by less than 0.5 percent although price increases 0.25 percent. This result suggests that maintaining price stability through petroleum price subsidies comes with a foregone output cost. Greater government's interventions in petroleum price fluctuations

manages more stable inflation but at the expense of larger output contraction.

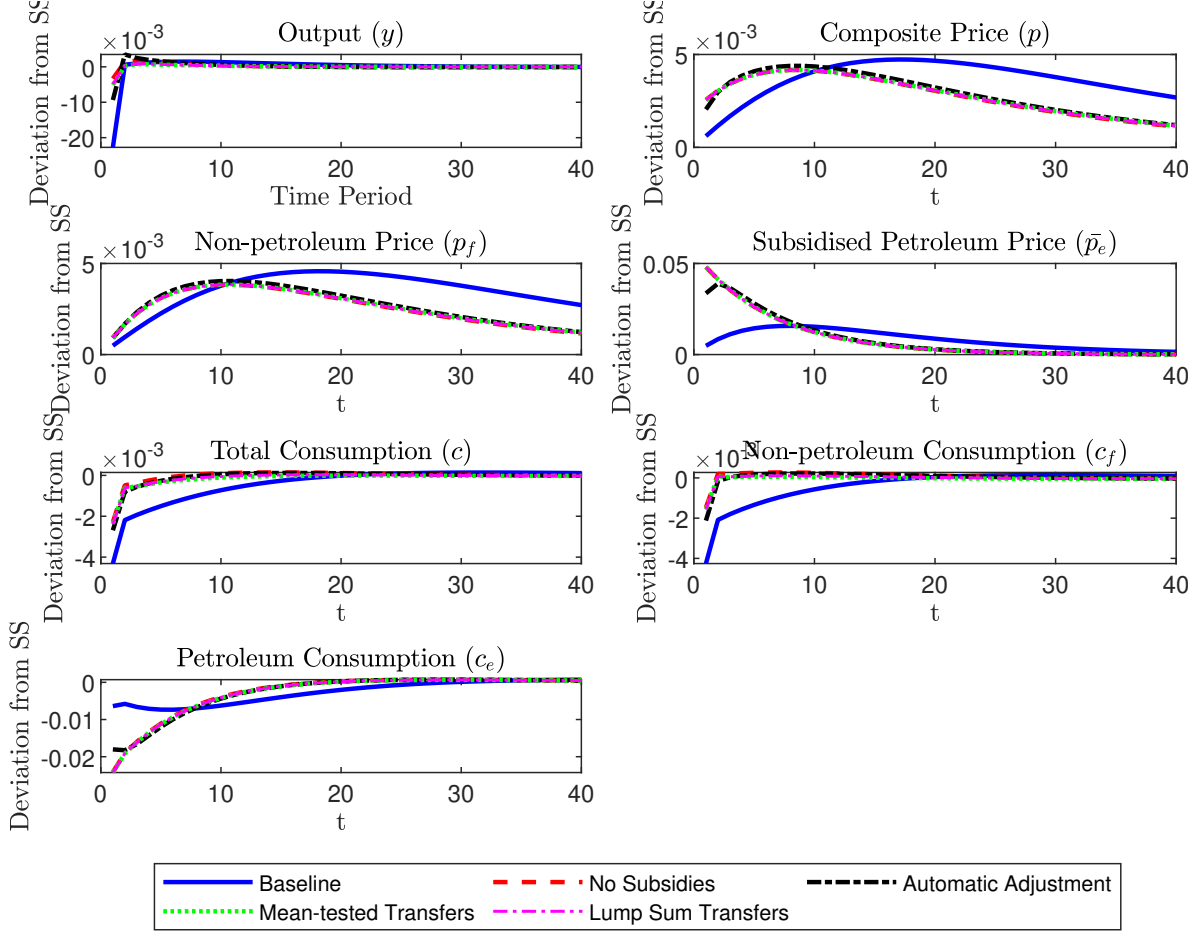


Figure 8: Impulse Response I

What explains such a phenomenon? In particular, why does the foregone output cost become larger as the government intervenes more on petroleum price fluctuations following the shocks? Why does stricter price control under baseline regime counterproductively lead higher and more persistent prices? I argue there are several transmissions could explain this phenomenon. Under baseline regime, price control distorts relative prices, leading to inefficient input choices and resource misallocation. As represented in Table (4), firms overuse petroleum input, and so labour due to input complementarities under this regime. Consequently, when oil price shocks occur, both petroleum and labour inputs contract more sharply, resulting a larger decline in output. In addition, contractions in labour demand brings downward pressure to real wage (see Figure 8). This in turn causes deeper reduction in income and households' consumptions (see

Figure 8).

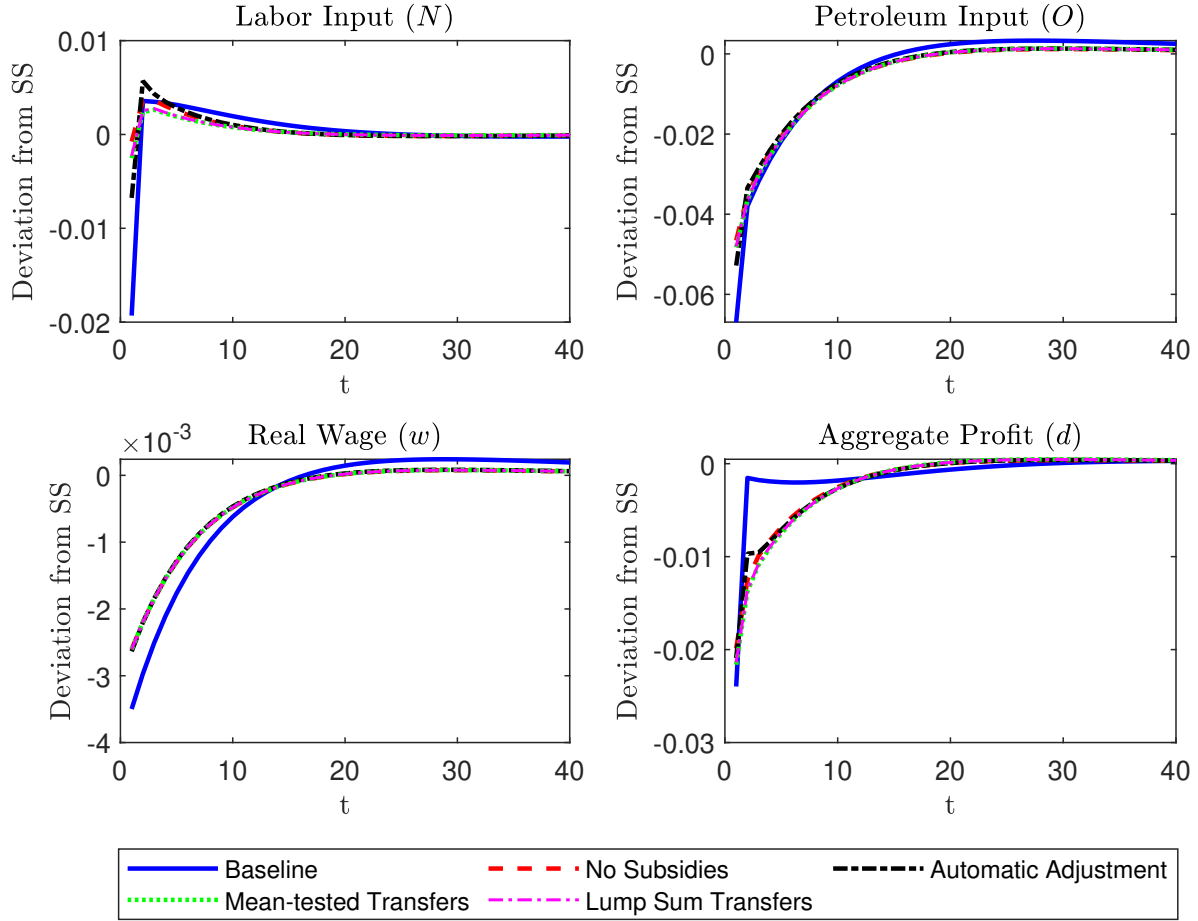


Figure 9: Impulse Response II

Now why do non-petroleum prices become higher for longer under the baseline regime? I argue there are two transmissions. First, the oil price shock raises firms' marginal costs through higher petroleum input prices, which puts upward pressure on non-petroleum prices. This applies not only for the baseline regime but also other regimes. However, the second mechanism differentiates the baseline from other regimes: Price subsidies distort households' expenditure patterns. Petroleum consumption declines only modestly relative to other regimes because households are cushioned from the increase in petroleum prices. In contrast, non-petroleum consumption drops more sharply and recovers for a longer of period. Both rising input costs and slower adjustments of petroleum consumption cause firms to set a more persistent increase in non-petroleum prices to maintain their profits. This result indicates the counterproductive consequence of petroleum price control on price level dynamics. This finding

strengthens Erceg et al. (2024) who find that energy subsidies may well be counterproductive. However, the different is that my results implicate domestic distortions while their arguments highlight external debt channel.

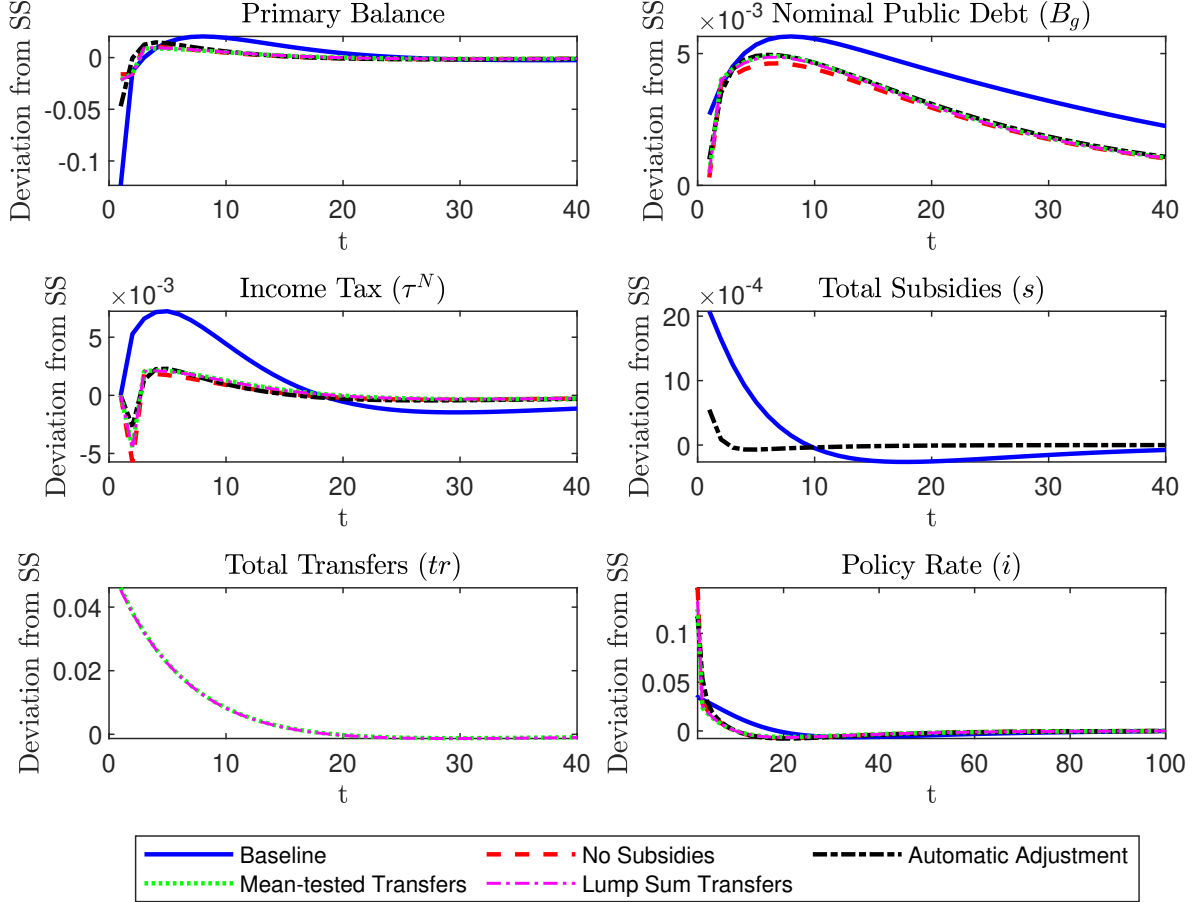


Figure 10: Impulse Response III

Finally, Figure (10) reports the implications on macroeconomic policy conditions, in particular fiscal policy. Following oil price shocks, primary balance deteriorates in all regimes. This is mainly because of tax revenue reduction as consumption and labour income are contracted. However, such a deterioration in primary balance is more prevalent under the baseline regime. Larger contractions in consumption and labour income, combined with increases in subsidies, lead to greater drop in primary balance in the baseline regime. This in turn causes larger and more persistent debt rises. To ensure long-run debt sustainability, the labour income tax rate adjusts in response to debt dynamics. Under the baseline regime, the tax rate fluctuates more strongly, generating additional distortions, especially in labour supply decisions. In addition,

policy rate also increases in response to inflation.

4.2.3 How Does Household Consumption Response Vary Across Wealth Levels?

This section addresses the question of who is most affected by oil price shocks. To answer this question, I identify the heterogeneity of household characteristics across wealth levels. In this regard, household income consists of several components: Net labour income $(1 - \tau_t^N)w_t e_t N_t$, return on assets $r_t b_t$, dividend from firms' profit d_t , and cash transfers tr_t . The relative importance of each component depends on the wealth level and policy regimes. For the households with low asset holdings, the majority of income derived from net labour income. As a result, **labour market channel** becomes particularly important. In contrast, the relative importance of $r_t b_t$ increases for the higher households with higher asset holdings, making **assets holding channel** crucial. Transfers tr also plays important role, especially under mean-tested transfers regime because transfer is decreasing in pre-tax income level. Low-income households benefit more than higher income households. In addition, d_t is equally distributed so the relative importance decreases as income is higher.

Figure (11) displays consumption responses to oil price shocks across asset holdings level and policy regimes. In general, households with lower asset holdings experience larger drop in consumption than high asset holdings households. Substantial reductions in labour input and real wages following the oil price shock disproportionately affect low-asset households. In contrast, for the high-asset households, the adverse effects of labour market channel are partly offset by the wealth effect associated with asset holdings channel. Specifically, although high-asset households also experience labour market contractions, increases in policy rate in response to petroleum-induced inflation raise returns on asset, and thus cushion the decline in income. Consequently, poorer households experience larger adverse impact on consumption than rich households due to stronger negative impact on income (Pieroni, 2023).

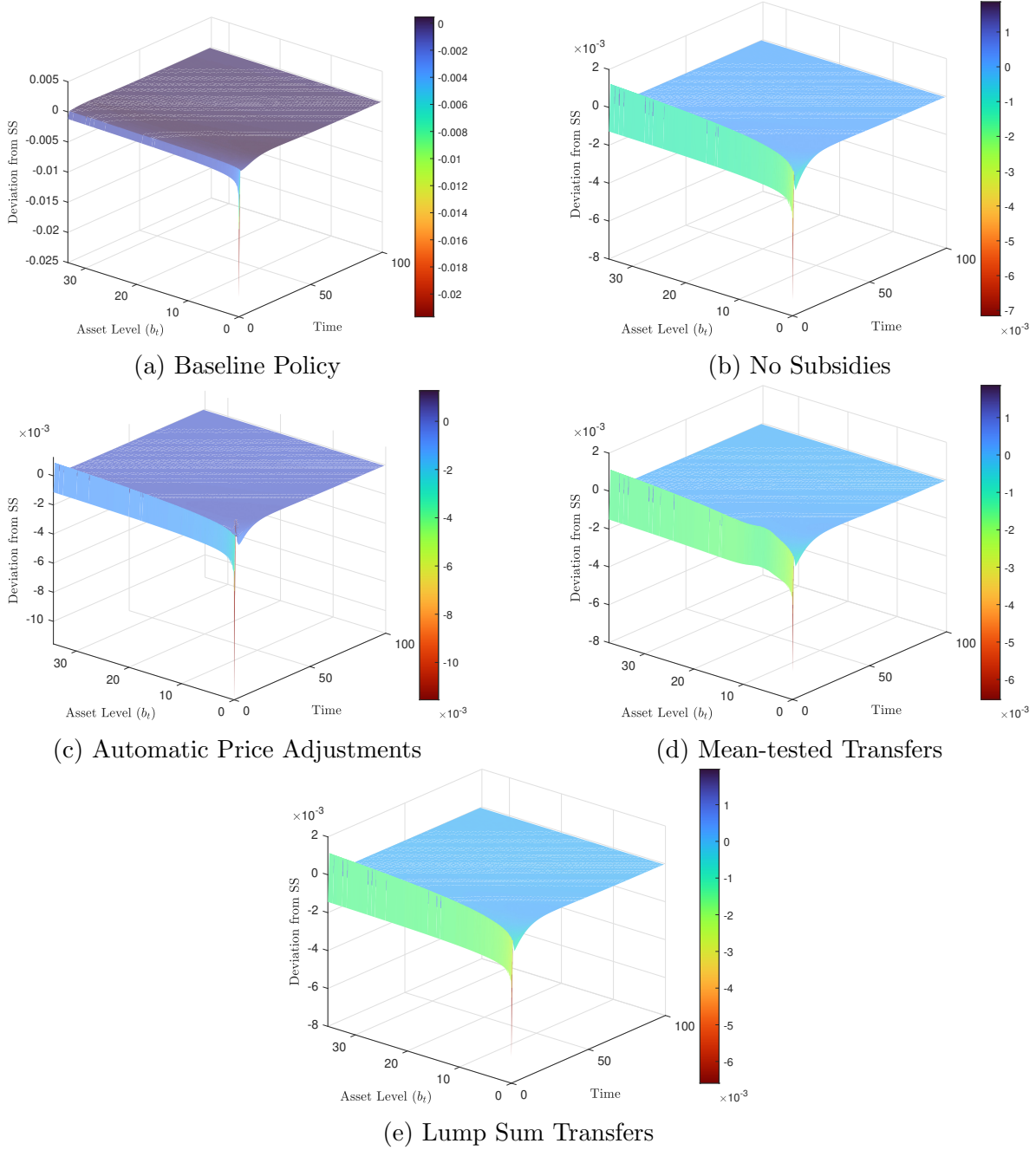


Figure 11: Consumptions Responses

Furthermore, the impacts are different across policy regimes. Under the baseline regime, the poorest experiences more than 2.1 percent decline in consumption, and this adverse effect is decreasing in asset holdings. This drop is the largest compared to other regimes. Consumption of the poorest household drops only by 1.1 percent in the automatic price adjustment regime, and around 0.6 percent in the no subsidy, mean-tested

transfers, and lump sum transfers regimes. This finding indicates that the baseline regime performs worse to protect the low-asset households from the shocks. Under the baseline regime, misallocation problem amplifies the labour market contractions, leading to larger income shock to the low-asset households. Moreover, the benefits of price control are regressive because high-asset households consume more petroleum than low-asset households. This further exacerbates the unequal effects among households.

5 Conclusion

Petroleum subsidies have been a long-standing tradition in Indonesia yet its effectiveness has been contested. This paper evaluates the capacity of current petroleum subsidies in Indonesia (***ad hoc* price adjustment** or baseline policy regime hereafter) by comparing it to alternative fiscal policy regimes including **no subsidies, automatic price adjustment, targeted transfers, and (universal) lump sum transfers**, along three dimensions. First, it evaluates the long run macroeconomic, fiscal, and equity-related implications. Second, this paper analyses the transitional consequences of subsidy reform. Lastly, this paper assesses which policy regimes can provide the best balance between macroeconomic stabilisation and fiscal sustainability.

This paper highlights several core findings. In the long run, removing all petroleum subsidies increases output, improves welfare, promotes a more equal distribution of income and wealth, and strengthens fiscal positions albeit at the expense of higher price level. Providing transfers can further improve welfare and equality in the long run, although they cause a slightly lower output compared to no subsidy regime. Despite these long run gains, this paper shows that reforms are challenging as they are associated with economic contraction and widening fiscal deficits during transitions.

As a stabilisation instrument, I find that baseline regime is counterproductive. Although it mitigates initial price surges, it also causes larger the output losses and more persistent inflation dynamics. This is primarily because of distortions arising from price control which affect relative prices and lead to inefficient input choices and resource misallocation. Furthermore, the baseline regime also performs worse to protect the poor from the shocks. Misallocation problem amplifies the labour market contractions, leading to larger income shock to the low-asset households. The benefits of price control are also regressive because high-asset households consume more petroleum than low-asset households. This further exacerbates the unequal effects among households. Primary balance also strongly deteriorates under the baseline regime. Larger contractions in consumption and labour income, combined with increases in subsidies, lead to greater drop in primary balance and more persistent debt rises.

This paper has several limitations that may provide avenues for future research. First, this paper abstracts external transmission channels such as foreign debt and exchange rate. As mentioned by Erceg et al. (2024), external channel plays an important role. Specifically, energy subsidies may affect macroeconomic outcomes through foreign debt deterioration, exchange rate depreciation, and therefore exchange pass-

through to domestic inflation.

Second, the HANK model in this paper uses canonical New Keynesian structure of labour market, i.e., nominal wage rigidity. Although it captures the labour market frictions, it does not actually account for the explicit number of unemployment. Incorporating with search-and-matching model (SAM) can enhance the analysis. For instance, Grancini et al. (2026) combines HANK with SAM and evaluate the effectiveness of income-stabilisation policies such as Unemployment Insurance (UI), targeted transfers, and lump sum transfers. Using the HANK-SAM, we may be able to explore, for instance, the role of UI in response to oil price shocks.

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A Appendices

A.1 Fuel Subsidies Revisited: Cross-countries Practices

Fuel subsidies are globally pervasive (Couharde and Mouhoud, 2020). Although many suggest that fuel subsidies should be abandoned, the oil price shocks in the 2020s prompted many countries to introduce, reintroduce, or expand such measures (World Bank, 2025). Black (2023) estimated that the total global subsidies in 2022 amounted to USD 7 trillion or equivalent to 7.1% of the GDP.⁹

Fuel subsidies, or government intervention in general, have a variety of forms: (i) regulated price caps; (ii) tax or excise reduction; (iii) fiscal transfer to oil companies; (iv) targeted subsidies to certain users; (v) price controls on upstream energy-related industries (World Bank, 2025). Table A.6 shows the implementation of government interventions related to controlling the domestic fuel price and containing the adverse impacts of energy price shocks in several countries. In Bahrain, Iran and Venezuela, the government sets the retail fuel prices below the market and covers the gap. These two countries **barely adjust the retail price**, although the international prices are fluctuated. However, there are several differences in the implementation. While there is no limitation in Bahrain, the Iranian government apply quotas for fuel every vehicle can purchase, and in Venezuela, only specific groups from population can access fuel at subsidised price.

Instead of keeping the price fixed for a long time, government in Bangladesh, Malaysia and China implements **automatic price adjustment** for subsidised fuel. More specifically, the price is adjusted frequently according to specific formula. In Brazil and the Philippines, the government has deregulated fuel prices, allowing the to fluctuate according to the market forces. However, the government still provide **some intervention to specific sectors**. In Brazil, the government delivers full fuel tax exemption and financial assistance to fishing vessels while in the Philippines the government gives 30% fuel price discount for farmers and fisher-folks and cash transfer to public transport drivers.

In developed economies, the fuel prices are deregulated but they conduct a variety of **unconventional/temporary fiscal measures** in response to the recent energy price shocks. In Norway, the Netherland and the United Kingdom, the government

⁹Estimated fuel subsidies include explicit and implicit subsidies. Explicit subsidies capture the gaps between subsidised and market-efficient price. Implicit subsidies consist of environmental and social costs.

provided fuel tax cuts; In Italy, the government provided fuel bonus of EUR 150 to aid low-income families who own a car and petrol coupons and tax exemptions for employers; The Spanish government implemented tax refund on commercial use of diesel in transportation sector.

There are several studies exploring how such measures may perform, especially during the price shocks. Ginn (2024), for instance, compared universal and targeted fuel subsidies in Malaysia in the presence of oil price shocks. They find that targeted subsidies is more favorable than universal subsidies as it is fiscally more stable while at the same time still being able to protect the vulnerable.

Motivated by the fiscal responses of energy price shocks in the Europe, Glocker and Wegmüller (2024) explore the role of various fiscal instruments to cushion the effects of energy price spikes. More specifically, they consider seven policy scenarios: (i) subsidies to energy-producing firms; (ii) subsidies to the energy expenses of goods producing firms; (iii) subsidies to households' energy consumption; (iv) labour and (v) general consumption taxes; (vi) income subsidies to particular household groups; and (vii) general government consumption spending. They find that fiscal policy conducted through subsidies/tax cuts on energy sales and on producer's expenditures on energy can neutralize energy price spikes, moderate inflation, and mitigate output loss. This, on the other hand, deteriorates public finances stability but only slightly.

In general, Dao et al. (2023) examined the macroeconomic economic consequences of the so-called unconventional fiscal measures in Euro Area during the post-pandemic shocks. They find that such measures can reduce one to two percentage points of inflation in 2022 and overcome overshoot afterwards. They also highlighted that those measures were can help to moderate long-term inflation expectation.

Furthermore, Erceg et al. (2024) find that, under closed economy framework, energy subsidies to consumer may be able to reduce core inflation. However, there is also potential counterproductive effect if the economy is open. If the energy shocks are persistent, it can deteriorate external debt, depreciate exchange rate, and thus trigger exchange rate pass-through. They suggest that if the objective to protect the poor, then targeted transfers are more efficient because it is less costly then universal subsidies.

Lastly, the seminal works done by Auclert et al. (2023) explore the role of monetary and fiscal policy in managing energy price shocks using heterogeneous-agent framework. For fiscal policy, they find that it can curtail the adverse effect of oil price shocks on GDP, and the easiest is to use price subsidies. Such measures insulate the households from energy price spikes, preserve real income and aggregate demand.

Table A.6: Fuel Subsidies in Several Countries, 2024

Country	Types of Subsidies	Explanation
Bahrain	Regulated Price (Ad Hoc Price Adjustments)	The government regulates the retail price of fuels and covers the gap between domestic retail price and international price. More specifically, retail price for gasoline and diesel have been frozen since 2019.
Bangladesh	Regulated Price (Automatic Price Adjustment)	Prior to 2024, the government set the retail price for fuels and it has been frozen for a quite long time. However, in March 2024, the government introduced automatic adjustment mechanism which is essentially adjust the subsidised price regularly on monthly basis.
Brazil	Targeted Subsidies	The authority has abandoned fuel subsidies and deregulated the price according to the market. However, the government is still providing intervention for fishing vessels. More specifically, the government provides full tax exemption and financial assistance equivalent to the gap between international and desired prices.
China	Regulated Price (Automatic Price Adjustments)	Chinese government has been constantly providing fuel price subsidies from the 1990s to ensure price stability and protect consumer's welfare. More specifically, the government sets price bands (ceiling and floor price) every 10 working days. The price set by the government takes into account several factors including international price, storage, transportation cost, and profit margin for the retailers.
Iran	Regulated Price (Ad Hoc Price Adjustments) with Quota	The government regulates the price retail price of fuels. The last price adjustment was undergone in 2019. For gasoline, the government implements quota for every vehicle: (i) 60 liters per month at price of 15,000 rials per liter (USD 0.36); and (ii) 300 liters per month for taxis and trucks. If the limit is exceeded, then the consumers pay at 30,000 rials per liter (USD 0.71)
Italy	Deregulated Price with Unconventional Measures during Shocks	The Italian government is essentially letting the fuel price fluctuates according to the market. However, in response to energy price shocks in 2022, the government provided several unconventional measures to cushion the adverse impact: (i) Fuel bonus of €150 to aid low-income families who own a car; and (ii) Petrol coupons and tax exemptions for employers.
Japan	Deregulated Price with Unconventional Measures during Shocks	Japanese government has deregulated fuel price since April 1996. However, due to energy price shocks in 2022, the government provided subsidies to domestic refiners and importers. The commodities included were gasoline, kerosene, diesel, fuel oil, jet fuel, and LPG.
Malaysia	Regulated Price with Partially Targeted Subsidies	The government regulates the fuel prices according to international price. The prices are adjusted regularly. For diesel, the government deregulated the price but providing financial assistance to logistic sectors.
Netherland	Deregulated Price with Unconventional Measures during Shocks	Fuel prices are deregulated but the government provided fuel tax reduction in response to energy price shocks in 2022.
Norway	Deregulated Price with Unconventional Measures during Shocks	Fuel prices are deregulated but the government provided fuel tax reduction in response to energy price shocks in 2022.
Spain	Deregulated Price with Unconventional Measures during Shocks	The fuel prices are deregulated. However, in response to the energy price shocks in 2022, the government implemented the partial tax refund on commercial use of diesel.
The Philip-pines	Targeted Subsidies	The government has deregulated the fuel prices since 1998. However, the government remains providing targeted fuel subsidies for certain groups: (i) Farmers and fisherfolks receive 30% discount ; and (ii) public transport drivers such as taxis, tricycles, and full-time ride-hailing and delivery services receive cash cards.
The UK	Deregulated Price with Unconventional Measures during Shocks	Fuel prices are deregulated but the government provided fuel tax reduction in response to energy price shocks in 2022.
Venezuela	Regulated Price with Targeted Subsidies	Since 2018, the government implements dual price regime: Subsidised and non-subsidie. The subsidised fuel prices have been frozen for a long time and provided to only certain groups with specific income level from population. This subsidies are integrated to broader social protection programs. For the non-subsidised fuel, the prices flutuate according to the market.

Source: World Bank (2025)

A.2 Model Derivation

A.2.1 Composite Price and Nominal Consumption

In this section, we will show how to derive nominal composite consumption $p_t c_t$ and composite price index p_t . In this model, there are two commodities consumed by households $i = \{f, e\}$, where f non-petroleum commodity and e is petroleum. We define the composite consumption as in equation (1).

Let us now solve the cost-minimization problem of total consumption by choosing c_f and c_e as the following Lagrangian form:

$$\mathcal{L}^{HH} = p_{f,t} c_{f,t} + \bar{p}_{e,t} c_{e,t} - \lambda_t \left[\left(\alpha^{\frac{1}{\sigma^h}} c_{f,t}^{\frac{\sigma^h}{\sigma^h-1}} + (1-\alpha)^{\frac{1}{\sigma^h}} c_{e,t}^{\frac{\sigma^h}{\sigma^h-1}} \right)^{\frac{\sigma^h-1}{\sigma^h}} - c_t \right] \quad (\text{A.14})$$

Theoretically, the Lagrangian multiplier can be interpreted as shadow price so $\lambda_t = p_t$. The First-order conditions from equation (A.14) imply the demand for each commodities:

$$\frac{p_{f,t}}{p_t} = \left(\alpha^{\frac{1}{\sigma^h}} c_{f,t}^{\frac{\sigma^h}{\sigma^h-1}} + (1-\alpha)^{\frac{1}{\sigma^h}} c_{e,t}^{\frac{\sigma^h}{\sigma^h-1}} \right)^{\frac{1}{1-\sigma^h}} \alpha^{\frac{1}{\sigma^h}} c_{f,t}^{-\frac{1}{\sigma^h}}$$

Where $\left(\alpha^{\frac{1}{\sigma^h}} c_{f,t}^{\frac{\sigma^h}{\sigma^h-1}} + (1-\alpha)^{\frac{1}{\sigma^h}} c_{e,t}^{\frac{\sigma^h}{\sigma^h-1}} \right) = c_t^{\frac{\sigma^h-1}{\sigma^h}}$ and solve for $c_{f,t}$ we thus obtain demand for non-petroleum consumption:

$$c_{f,t} = \alpha \left(\frac{p_{f,t}}{p_t} \right)^{-\sigma^h} c_t \quad (\text{A.15})$$

We can do the same process for $c_{e,t}$ and obtain the demand for petroleum commodity as the following

$$c_{e,t} = (1-\alpha) \left(\frac{\bar{p}_{e,t}}{p_t} \right)^{-\sigma^h} c_t \quad (\text{A.16})$$

By substituting equations (A.15) and (A.16) to (1), we can solve for the composite price as follows:

$$c_t = \left[c_t^{\frac{\sigma^h-1}{\sigma^h}} \left(\alpha \left(\frac{p_{f,t}}{p_t} \right)^{-\sigma^h} + (1-\alpha) \left(\frac{\bar{p}_{e,t}}{p_t} \right)^{-\sigma^h} \right) \right]^{\frac{\sigma^h}{1-\sigma^h}}$$

Thus,

$$p_t = \left(\alpha p_{f,t}^{1-\sigma^h} + (1-\alpha) \bar{p}_{e,t}^{1-\sigma^h} \right)^{\frac{1}{1-\sigma^h}} \quad (\text{A.17})$$

From the equations above, now we can derive the total nominal consumption as the following:

$$\begin{aligned} p_{f,t} c_{f,t} + \bar{p}_{e,t} c_{e,t} &= \alpha \left(\frac{p_{f,t}}{p_t} \right)^{-\sigma^h} c_t p_{f,t} + (1-\alpha) \left(\frac{\bar{p}_{e,t}}{p_t} \right)^{-\sigma^h} c_t \bar{p}_{e,t} \\ \iff p_{f,t} c_{f,t} + \bar{p}_{e,t} c_{e,t} &= c_t \left[p_t^{\sigma^h} \left(\alpha p_{f,t}^{1-\sigma^h} + (1-\alpha) \bar{p}_{e,t}^{1-\sigma^h} \right) \right] \end{aligned}$$

Then,

$$p_{f,t} c_{f,t} + \bar{p}_{e,t} c_{e,t} = c_t \left(p_t^{\sigma^h} p_t^{1-\sigma^h} \right) = c_t p_t \quad (\text{A.18})$$

Equation (A.18) proves what is elaborated on footnote (4).

A.2.2 Wage New Keynesian Philips Curve

For the supply side of the labour market, there two primary agents namely recruiting firm and union. Recruiting firm demands labour from different unions in perfectly competitive market, and the objective is to optimize the profit:

$$\max_{N_t, N_t(u)} W_t N_t - \int_0^1 W_t(u) N_t(u) du \quad \text{s.t.} \quad N_t = \left[\int_0^1 N_t(u)^{\frac{\theta^W}{\theta^W-1}} du \right]^{\frac{\theta^W-1}{\theta^W}} \quad (\text{A.19})$$

where W_t is nominal wage.

The first-order condition implies demand for labour from recruiting firm as follows:

$$N_t(u) = \left(\frac{W_t(u)}{W_t} \right)^{-\theta^W} N_t \quad (\text{A.20})$$

Demand for labour from recruiting firms become the constraint of the optimization problem for the union (see in Section 3.2.1). The first-order condition of the union problem with respect to nominal wage every unions set $W_t(u)$ is therefore

$$\begin{aligned} (1-\theta^W) W_t(u)^{-\theta^W} \frac{W_t^{\theta^W} N_t}{p_t} + \theta^W \frac{N_t^\psi}{C_t^{1-\gamma}} W_t(u)^{(-\theta^W-1)} \\ - \Psi^W \left[\frac{W_t(u)}{W_{t-1}(u)} - 1 \right] \frac{N_t}{W_{t-1}(u)} - \beta \Psi^W \mathbb{E}_t \left[\left(\frac{W_{t+1}(u)}{W_t(u)} - 1 \right) \left(-\frac{W_{t+1}(u)}{W_t(u)} \right) N_{t+1} \right] = 0 \end{aligned} \quad (\text{A.21})$$

By symmetric equilibrium $W_t = W_t(u)$ and some manipulations (i.e., dividing both

sides with N_t and multiplying both sides with W_t), we get the wage NKPC as in equation (3).

A.2.3 Inflation New Keynesian Philips Curve

In this section, we simply elaborate on the Lagrangian form of intermediate firm problem and the first-order condition to show how we obtain inflation NKPC (see equation 6). The following equation is the Lagrangian form:

$$\mathcal{L}^F = \mathbb{E}_t \sum_{s=0}^{\infty} \Lambda_{t,t+s} p_{t+s} \left[\begin{aligned} & \frac{p_{t+s}(f)Y_{t+s}(f)}{p_{t+s}} - w_{t+s}N_{t+s}(f) - \frac{p_{e,t+s}}{p_{t+s}} O_{t+s}(f) \\ & - \frac{\Psi^F}{2} \left(\frac{p_{t+s}(f)}{p_{t+s-1}(f)} - 1 \right)^2 Y_{t+s} \\ & + mc_{t+s}(f) \left(A_t N_t^{\alpha^F} O_t^{(1-\alpha^F)} - Y_t(f) \right) \end{aligned} \right] \quad (\text{A.22})$$

The first-order condition with respect to $N_t(j)$ $O_t(f)$ will result in demand for labour and petroleum inputs, respectively. Furthermore, the first-order condition with respect to $p_t(f)$

$$\begin{aligned} & \left[(1 - \theta^F) \left(\frac{p_t(f)}{p_{f,t}} \right)^{-\theta^F} Y_t \right] - \left[Y_t p_t \Psi^F \left(\frac{p_t(f)}{p_{t-1}(f)} - 1 \right) \left(\frac{1}{p_{t-1}(f)} \right) \right] \\ & + \left[p_t mc_t(f) (-\theta^F) p_t(f)^{(-\theta^F-1)} p_{f,t}^{-\theta^F} Y_t \right] \\ & + \mathbb{E}_t \left[\Lambda_t \left(\frac{p_{t+1}(f)}{p_t(f)} - 1 \right) Y_{t+1} p_{t+1} \left(\frac{p_{t+1}(f)}{p_t(f)^2} \right) \right] = 0 \end{aligned} \quad (\text{A.23})$$

By symmetric equilibrium $p_t(f) = p_{f,t}$ and manipulations including dividing by p_t and Y_t and multiplying by $p_{f,t}$, we can obtain inflation NKPC as in equation (6).

A.3 Competitive Equilibrium

The competitive equilibrium is defined as sequence of (a) distributions over households states $\{\Gamma_t\}_{t=0}^\infty$; (b) households' value function $\{V_t, g_{c,t}, g_{b,t}\}_{t=0}^\infty$; and (c) aggregate quantities and prices $\{C_t, C_{e,t}, C_{f,t}, Y_t, N_t, O_t, b_t^g, \tau^N, G_t, s_t, d_t, tr_t, i_t, r_t, w_t, p_t, p_{f,t}, \bar{p}_{e,t}, \pi_t, \pi_t^W, \pi_t^F\}_{t=0}^\infty$ such that

$$\left\{ \begin{array}{l} Y_t = A_t N_t^{\alpha_F} O_t^{(1-\alpha_F)} \\ N_t = \alpha_F \left(\frac{mc_t}{w_t} \right) Y_t \\ O_t = (1 - \alpha_F) \left(\frac{p_t mc_t}{p_{e,t}} \right) Y_t \\ p_t = \left(\alpha p_{f,t}^{1-\sigma^h} + (1 - \alpha) \bar{p}_{e,t}^{1-\sigma^h} \right)^{\frac{1}{1-\sigma^h}} \\ d_t = \left(\frac{p_{f,t}}{p_t} \right) Y_t - w_t N_t - \left(\frac{p_{e,t}}{p_t} \right) O_t - \Phi_t^W \\ \pi_t^F = \frac{p_{f,t}}{p_{f,t-1}} \\ \pi_t^F (1 + \pi_t^F) = \kappa^F (mc_t - \mu^F \frac{p_{f,t}}{p_t}) + \Lambda_t \mathbb{E}_t \left[\pi_{t+1}^F (1 + \pi_{t+1}^F) (1 + \pi_{t+1}) \frac{Y_{t+1}}{Y_t} \right] \\ \pi_t^W = \frac{W_t}{W_{t-1}} \\ \Psi^W \pi_t^W (1 + \pi_t^W) = \kappa^W \left(\frac{N_t^\psi}{C_t^{-\gamma}} - \mu^W w_t \right) + \beta \mathbb{E}_t \left[\pi_{t+1}^W (1 + \pi_{t+1}^W) \frac{N_{t+1}}{N_t} \right] \\ i_t = r_{ss} + \phi_\pi \pi_t \\ 1 + r_t = \frac{1+i_{t-1}}{1+\pi_t} \\ \tau_t^N w_t N_t + \tau_t^c c_t + b_t^g = (1 + r_t) b_{t-1}^g + G_t + tr_t + s_t \\ s_t = \frac{\bar{p}_{e,t} - p_{e,t}}{p_t} \left(\int c_{e,t} d\Gamma_t \right) \\ \tau_t^N = \tau_{ss}^N + \iota_b (b_{t-1}^g - b_{ss}^g) \\ \bar{p}_{e,t} = \omega \bar{p}_{e,t-1} + (1 - \omega) \varphi p_{e,t} \\ tr_{h,t} = \left(\frac{\varphi^{tr}}{(1 + \kappa^{tr} \exp(y_{h,t} - \bar{y}))} \right) \frac{p_{e,t}}{p_t} \int c_{e,t} d\Gamma_t \\ C_t = \int g_{c,t}(b, e) d\Gamma_t \\ C_{f,t} = \alpha \left(\frac{p_{f,t}}{p_t} \right)^{-\sigma^h} C_t \\ C_{e,t} = (1 - \alpha) \left(\frac{\bar{p}_{e,t}}{p_t} \right)^{-\sigma^h} C_t \\ b_t^g = \int g_{b,t}(b, e) d\Gamma_t \\ Y_t = C_{f,t} + G_t + \Phi_t^p + \Phi_t^W \\ \Phi_t^p = \frac{\Psi^W}{2} \left(\frac{W_t(u)}{W_{t-1}(u)} - 1 \right)^2 N_t \\ \Phi_t^W = \frac{\Psi^F}{2} \left(\frac{p_t(f)}{p_{t-1}(f)} - 1 \right)^2 Y_t \end{array} \right.$$

A.4 Computational Method

A.4.1 Rouwenhorst Discretisations

Idiosyncratic shocks defined in equation (2) follow AR(1) process in log-form.

$$\log(e_{t+1}) = \rho_e \log(e_t) + \varepsilon_{e,t+1}, \quad \varepsilon_e \sim N(0, \sigma_\varepsilon)$$

Note that this process has the following mean and variance properties.

$$\mathbb{E}(e) = 0, \quad \sigma_e^2 = \frac{1}{1 - \rho_e^2} \sigma_\varepsilon^2$$

The Rouwenhorst discretisations have two main components, i.e., a set of grid points $\mathcal{E}$ and Markov chain transition matrix Π . Step one, we set an equally spaced, ordered grid $\mathcal{E} = \{\bar{e}_1, \dots, \bar{e}_N\}$ where $\bar{e}_1 = \sigma_e \sqrt{N-1}$ and $\bar{e}_N = -\sigma_e \sqrt{N-1}$. The second step is to construct Markov chain transition matrix. For $N = 2$,

$$\Pi_{N=2} = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$

where $p = (1 - \rho_e)/2$. For $N \geq 3$,

$$\Pi_{N \geq 3} = p \begin{bmatrix} \Pi_{N-1} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix} + (1-p) \begin{bmatrix} \mathbf{0} & \Pi_{N-1} \\ 0 & \mathbf{0} \end{bmatrix} + p \begin{bmatrix} \mathbf{0}' & 0 \\ \Pi_{N-1} & \mathbf{0}' \end{bmatrix} + (1-p) \begin{bmatrix} 0 & \mathbf{0}' \\ \mathbf{0}' & \Pi_{N-1} \end{bmatrix}$$

$\mathbf{0}'$ is $(N-1) \times 1$ zero matrix.

A.4.2 The Algorithm

Let $\mathbb{Z} \subseteq \mathbb{R}^{N_z}$ be the space of exogenous aggregate variables where N_z is number of aggregate variables, $\mathbb{Y} \subseteq \mathbb{R}^{N_y}$ denotes the space of endogenous aggregate variables and N_y is the number of variables, $\mathbb{E} \subseteq \mathbb{R}^{N_e}$ be the space of individual exogenous states, $\mathbb{A} \subseteq \mathbb{R}^{N_a}$ represents the individual endogenous states and N_a is the number of states, and lastly $\mathbb{X} \subseteq \mathbb{R}^{N_x}$ denotes individual decision rule where N_x is the number of individual choice variables.

With a sufficiently large time period T , the equilibrium is characterised by system of equations for endogenous aggregate variables $\mathcal{Y} \equiv \{Y_t\}_{t=0}^T$ as follows.

$$\mathcal{F}_t(\{Z_s\}_{s=0}^t, \{Y_s\}_{s=0}^t, \bar{x}_t(\mathcal{Z}, \mathcal{Y})) = 0, \quad \text{for } t = 0, 1, \dots, T \quad (\text{A.24})$$

where $\mathcal{F}_t : \mathbb{Z}^{T+1} \times \mathbb{Y}^{T+1} \times \mathbb{X} \rightarrow \mathbb{R}^{N_y}$, $\mathcal{Z} \equiv \{Z_t\}_{t=0}^T$ is a sequence of exogenous aggregate variables, and $\bar{x}_t(\mathcal{Z}, \mathcal{Y})$ is aggregate individual decision rule. Specifically,

$$\bar{x}_t(\mathcal{Z}, \mathcal{Y}) = \int g_{x,t}(e, a; \mathcal{Z}, \mathcal{Y}) \, d\Gamma_t$$

where Γ_t is individual states distribution at time t and $g_{x,t}(e, a; \mathcal{Z}, \mathcal{Y})$ is decision rule derived from individual optimality condition. We can obtain the decision rule $g_{x,t}(e, a; \mathcal{Z}, \mathcal{Y})$ using Value Function Iteration (VFI).

System equations expressed in (A.24) can be solved using multivariate linear approximation. Let $\bar{\mathcal{Y}}_t \equiv \{Y_s\}_{s=0}^t$,

$$\bar{\mathcal{Y}}_t^{(n+1)} = \bar{\mathcal{Y}}_t^{(n)} - \mathbf{J}(\{Z_s\}_{s=0}^t, \{Y_s\}_{s=0}^t, \bar{x}_t(\mathcal{Z}, \mathcal{Y}))^{-1} \mathcal{F}_t(\{Z_s\}_{s=0}^t, \{Y_s\}_{s=0}^t, \bar{x}_t(\mathcal{Z}, \mathcal{Y}))$$

where $n \in \mathbb{N} \cup \{0\}$ is iteration step and $\mathbf{J}(\{Z_s\}_{s=0}^t, \{Y_s\}_{s=0}^t, \bar{x}_t(\mathcal{Z}, \mathcal{Y}))$ is the Jacobian matrix (i.e., first-order condition of $\mathcal{F}_t$ with respect to $\bar{\mathcal{Y}}_t$). The real bottleneck to solve (A.24) is to calculate the Jacobian matrix $\mathbf{J}$ efficiently. In this paper, I use Broyden Updates method to calculate the Jacobian matrix. The algorithm is as the following:

1. Guess $\bar{\mathcal{Y}}_t^{(0)}$ and calculate $\mathbf{J}^{(0)}$ and $\mathcal{F}^{(0)}$.
2. Update the Jacobian using the following formula:

$$\mathbf{J}^{(n)} = \mathbf{J}^{(n-1)} + \frac{\Delta \mathcal{F}^{(n)} - \mathbf{J}^{(n-1)} \Delta \bar{\mathcal{Y}}^{(n)}}{\|\Delta \bar{\mathcal{Y}}^{(n)}\|} \left(\Delta \bar{\mathcal{Y}}^{(n)} \right)', \text{ for } n > 0$$

where $\Delta \mathcal{F} = (\mathcal{F}^{(n)} - \mathcal{F}^{(n-1)})$ and $\Delta \bar{\mathcal{Y}} = (\bar{\mathcal{Y}}^{(n)} - \bar{\mathcal{Y}}^{(n-1)})$.

3. Update guess $\bar{\mathcal{Y}}^{(n+1)} = \bar{\mathcal{Y}}^{(n)} - (\mathbf{J}^{(n)})^{-1} \mathcal{F}^{(n)}$.
4. Stop iteration in step (2) and (3) if $\|\bar{\mathcal{Y}}_t^{(n+1)} - \bar{\mathcal{Y}}_t^{(n)}\| < \epsilon$ for $\epsilon > 0$.